Technical report · working draft, October 12, 2026

Rhizome Neural Operator

Technical Report: a state-dependent integral operator for 21 cm reionization fields

Haris Kapetanovic

Abstract

Neural operators such as the Fourier Neural Operator (FNO) learn maps between function spaces efficiently, but their kernels are global, stationary and band-limited, which makes sharp, localized structure such as the ionization fronts of the epoch of reionization hard to represent. Multi-scale hybrids like the U-Net and U-FNO recover local detail at the cost of resolution independence, and still use a fixed communication pattern. This report introduces the *rhizome neural operator*, a recurrent integral operator whose connections depend on the state of the field. A learned kernel, evaluated with the FFT, is modulated at every update by per-cell source and receiver gates computed from the current hidden state, so that the effective kernel is state-dependent at the cost of a single Fourier layer. On 21 `cmFAST` lightcones, the rhizome outperforms the best models of an existing FNO-based pipeline on every task tested: 3-D $x_{\rm HI}$ (test RMSE 0.059 against 0.068) and joint $x_{\rm HI}$ and 21 cm brightness temperature (0.050 / 5.8 mK against 0.056 / 6.5 mK). Its margin is largest on physical observables. A 2-D variant that sees ±32 cells of line-of-sight context and is applied slice by slice beats the native 3-D models on the same data. It halves the error in bubble sizes and in the field-level bias, sharpens ionization fronts from 6.2 to 3.9 Mpc, and predicts a full cone in under 6 s. A small-scale power term in the loss removes its one weakness, excess T_b structure before reionization.

Keywords: neural operators, Fourier neural operator, operator learning, 21 cm cosmology, epoch of reionization, emulation

CONTENTS

1 Neural Operators and FNOs **1**
1.1 Operator learning: problem setup . . . 1
1.2 Neural operators as stacked kernel integrals . . . 2
1.3 The Fourier Neural Operator . . . 2
1.4 Discretisation invariance and its limits . . . 2
1.5 Limitations of a purely spectral kernel . . . 3

2 Spatial structure in neural operators and the U-Net **3**
2.1 Inductive biases: locality, scale and stationarity . . . 3
2.2 The U-Net . . . 4
2.3 Hybrid architectures: U-FNO and related . . . 4
2.4 Open problem: structure that adapts to the field . . . 5

3 Rhizome Neural Operator approach **5**
3.1 Design principles . . . 5
3.2 Mathematical background: from kernel integrals to the rhizome . . . 6
3.3 State-dependent integral interaction . . . 7
3.4 Gating: source and receiver factors . . . 7
3.5 Recurrent time-stepping . . . 8
3.6 Discretization and spectral evaluation . . . 9
3.7 Model variants . . . 9
3.8 Assumptions and limitations . . . 10
3.9 Relation to FNO, U-FNO and attention . . . 10

4 Training and Results **10**
4.1 Datasets . . . 10
4.2 Training protocol . . . 11
4.3 Evaluation metrics . . . 11
4.4 Baselines . . . 11
4.5 2-D slices: line-of-sight context dominates . . . 12
4.6 3-D windows: $x_{\rm HI}$ . . . 12
4.7 3-D multi-field: inputs, heads and losses . . . 13
4.8 Slice-wise reconstruction beats native 3-D . . . 13
4.9 Best model: multi-field slice-wise rhizome . . . 13
4.10 Brightness temperature before reionization . . . 14
4.11 Learned kernels: no spectral collapse . . . 18
4.12 Cost . . . 18
4.13 Ablations . . . 18
4.14 Discussion . . . 18

A Notation **20**

1 NEURAL OPERATORS AND FNOS

1.1 Operator learning: problem setup

Neural Operators work by characterizing the inputs and outputs of its mapping as elements of function spaces. Let $f \in \mathcal{U}, g \in \mathcal{V}$ be functions defined on domains $D \subset \mathbb{R}^d$ and $D' \subset \mathbb{R}^{d'}$, respectively. The goal is to learn a mapping $G : U \to V$ such that $g = G(f)$. In practice, we observe these functions on discrete grids D_n and D'_m, leading to finite-dimensional representations of the functions.

The function space approach enables us to think of the mapping G as a continuous operator, suitable for the application of elements of functional analysis. This perspective allows for the design of architectures that are invariant to the discretization of the input and output functions, making them suitable for tasks where the underlying functions may be sampled at different resolutions. For example, in the context of solving PDEs, the input function f could represent initial conditions or parameters of the PDE, while the output function g could represent the solution of the PDE at a later time. The neural operator aims to learn this mapping in a way that is robust to changes in the discretization of the input and output functions.

1.2 Neural operators as stacked kernel integrals

One way of constructing a neural operator is with the approach of the Green's function of a linear operator. The Green's function $G(x, y)$ is the kernel of the integral operator that maps an input function f to an output function g via the integral

$$g(r) = \int_D \kappa(r, r', \theta) f(r')\, dr'. \tag{1}$$

With some external parameters θ. The choice of kernel $\kappa(r, r', \theta)$ determines the properties of the operator. A common approach is to assume that κ is translation invariant, meaning that

$$\kappa(r, r', \theta) = \kappa(r - r', \theta). \tag{2}$$

This assumption turns the operator effectively into a convolution, which can be efficiently computed in Fourier space. The kernel κ can be parameterized by a neural network, allowing the operator to learn complex mappings from input to output functions.

1.3 The Fourier Neural Operator

Since the kernel integral operator is translation invariant, it can be efficiently computed in Fourier space:

$$g(r) = \int_D \kappa(r - r', \theta) f(r')\, dr' = \mathscr{F}^{-1}\left[\mathscr{F}[\kappa] \cdot \mathscr{F}[f]\right](r), \tag{3}$$

The Fourier Neural Operator (FNO) leverages this approach in discrete function spaces, where the integral turns into a sum over discrete grid points:

$$g(r_i) = \sum_{j=1}^{N} \kappa(r_i - r_j, \theta) f(r_j) \Delta r_j, \tag{4}$$

where r_i and r_j are points on the discrete grid, and Δr_j is the grid spacing. The FNO computes this sum efficiently by transforming the input function f into Fourier space, applying the learned kernel κ in Fourier space, and then transforming back to physical space. This is achieved using the Fast Fourier Transform (FFT), which has a computational complexity of $\mathcal{O}(N \log N)$, making it suitable for large-scale problems. A diagram of the FNO architecture is shown in Figure 1. The FNO consists of three main components: a lifting layer that maps the input function f to a higher-dimensional representation, a series of Fourier layers that apply the learned kernel in Fourier space, and a projection layer that maps the output back to the desired function space. The lifting and projection layers are typically implemented as fully connected neural networks, while the Fourier layers consist of FFTs, learned multiplications in Fourier space, and inverse FFTs.

While it is useful to think about FNOs from this theoretical framework of kernel integral operators in PDEs, there is no obstacle in removing that context and examining the architecture as a general-purpose operator learning framework. What happens on paper is that the input data is transformed into Fourier space, where a learned kernel is applied. The learned weights correspond to the Fourier coefficients of the kernel, so instead of spatial context, the FNO is learning a global, stationary, spectral representation of the data. This is a central tradeoff of the FNO architecture: it is very efficient and can learn global structure, but it is limited in its ability to learn local, multi-scale features.

1.4 Discretisation invariance and its limits

In practice, the FNO does not learn the kernel κ directly. Each Fourier layer stores the Fourier coefficients of the kernel for a fixed set of low wavenumbers and sets all others to zero:

$$\hat{g}(k) = \begin{cases} R_\phi(k)\, \hat{f}(k), & |k| \le k_{\max}, \\ 0, & |k| > k_{\max}, \end{cases} \tag{5}$$

where $\hat{f} = \mathscr{F}[f]$, and the $R_\phi(k) \in \mathbb{C}^{d_v \times d_v}$ are learned complex matrices that mix the d_v channels of the lifted representation. The parameters therefore live on wavenumbers, not on grid points: their number is set by $k_{\max}$ and d_v and is independent of the number of grid points N. The same weights can be applied to an input sampled on any grid fine enough to resolve the retained modes, i.e. as long as the Nyquist wavenumber of the grid exceeds $k_{\max}$. The remaining components, namely the lifting P, the projection Q and the pointwise linear term W, act on each grid point separately and are likewise independent of the resolution. This is the sense in which the FNO is discretization invariant, and it enables the *zero-shot super-resolution* reported by Li et al. [6]: a model trained on a coarse grid can be evaluated on a finer grid without retraining.

This invariance is a property of the architecture, not of the learned map, and in practice it holds only approximately. Three caveats are relevant for this work.

Band-limited output. Since the spectral path only acts on $|k| \le k_{\max}$, evaluating a trained FNO on a finer grid does not add information on the newly resolved scales. Any structure above $k_{\max}$ in the output has to be generated by the pointwise nonlinearity σ acting on a band-limited signal. Super-resolution with an FNO is therefore closer to a learned, nonlinear interpolation than to a prediction of genuinely new small-scale structure.

Aliasing. The nonlinearity σ creates harmonics above the band of its input. On a continuous domain these would simply be additional high-k modes; on a grid, everything above the Nyquist wavenumber is folded back onto lower wavenumbers. Which modes are folded where depends on N, so the same network evaluated on two different grids gives slightly different outputs, even on the shared low-k band. The discrete network is thus not an exact discretization of a single continuous operator. This aliasing error has been analysed systematically by Bartolucci et al. [2].

Resolution-dependent targets. Even a perfectly discretization invariant architecture cannot compensate for the fact that the data themselves change with resolution. If the training fields come from a simulation at a given cell size, the smoothing scales, sub-grid prescriptions and sharpness of features are tied to that cell size. For the ionization fields considered here, bubble boundaries are resolved down to the grid scale, so a field simulated at higher resolution contains genuinely different small-scale structure that a model trained at lower resolution has never seen.

Finally, the FFT assumes periodic boundary conditions. This is natural for coeval simulation boxes, which are periodic by construction, but not for lightcones, whose line-of-

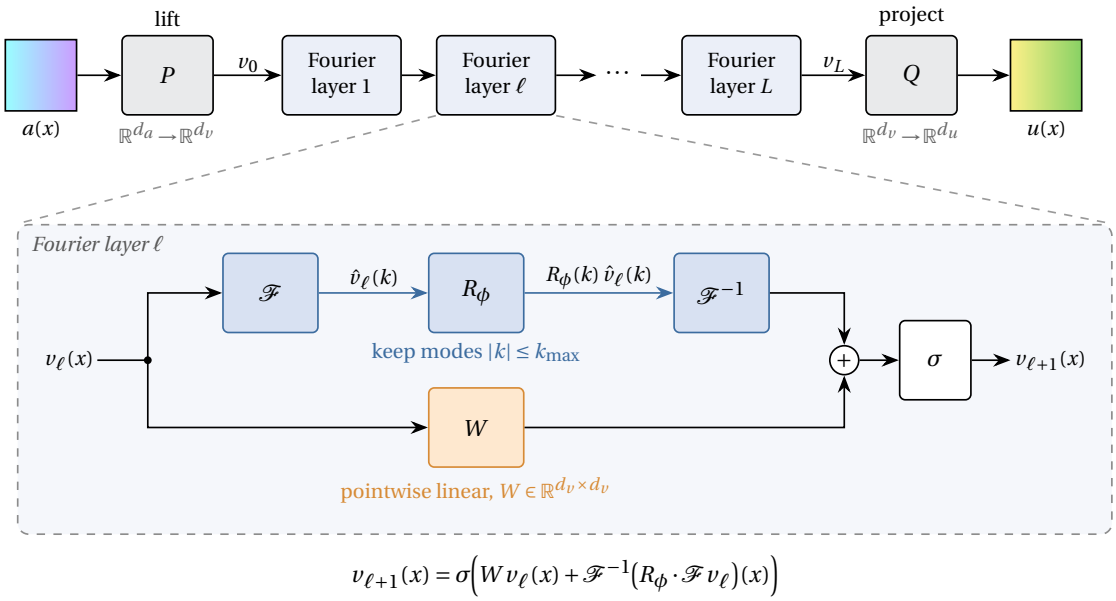


Figure 1. Fourier Neural Operator architecture.

sight direction is not periodic. There, padding or windowing is needed; without it, the convolution wraps around and couples the two ends of the lightcone.

1.5 Limitations of a purely spectral kernel

The end of Section 1.3 identified the central tradeoff of the FNO: its kernel is global, stationary and spectral. Each of these properties is a strength for smooth, statistically homogeneous fields, and each becomes a limitation for fields with sharp, localized structure such as ionization maps.

Stationarity. Because κ depends only on $r - r'$, the spectral path applies the same linear response at every position. How strongly a point r' influences a point r cannot depend on the state of the field at either location. All state dependence has to enter through the pointwise terms and the nonlinearity σ, and builds up only by stacking layers. Many of the processes we want to emulate are not of this form: in reionization, whether ionizing photons from a source reach a given region depends on whether the gas in between is already ionized.

Mode truncation and sharp features. The truncation in Equation (5) limits each spectral path to wavelengths longer than roughly $L/k_{\max}$, where L is the box size. Sharp features have broad spectra: the Fourier coefficients of a step decay only as $1/k$, so much of the information that locates an edge sits at high wavenumbers. Truncating these modes produces the familiar smoothing and Gibbs-type ringing around discontinuities. The ionization field in reionization simulations is close to binary, with cells either neutral or ionized, so its bubble boundaries are exactly this kind of feature. The FNO has to reconstruct them from the pointwise path W and the nonlinearity, which only see a single grid cell.

No intermediate scales. The spectral path is global and the pointwise path is local to one cell, but nothing in between is represented explicitly. Every Fourier layer operates on the full grid with the same set of modes; there is no coarse-to-fine hierarchy and no local receptive field spanning a few cells. Structures at intermediate scales, such as individual bubbles a few to a few tens of cells across, have to be represented as combinations of global modes.

Cost of more modes. The obvious remedy, increasing $k_{\max}$, is expensive. Each retained mode carries a full $d_v \times d_v$ complex matrix, so the number of spectral parameters grows as $\mathcal{O}(k_{\max}^d\, d_v^2)$, i.e. cubically in $k_{\max}$ for three-dimensional fields. Keeping all modes also removes the implicit regularization that truncation provides.

These limitations concern not the accuracy of the FNO on large scales, where it typically performs well, but the fidelity of small-scale, phase-sensitive structure. Such errors can be hidden in the power spectrum, which discards phase information, and become visible in the cross-correlation coefficient $r(k)$ between prediction and truth (Equation (20)). They motivate architectures that build spatial structure into the operator explicitly, which are the subject of Section 2.

2 SPATIAL STRUCTURE IN NEURAL OPERATORS AND THE U-NET

Section 1.5 showed that the FNO represents a field through global, stationary, band-limited modes plus a pointwise map. By *spatial structure* we mean everything this leaves out: locality (interactions that are strong between nearby points and weak between distant ones), an explicit hierarchy of scales, and interactions whose strength depends on where in the field they happen. Reionization fields need all three. Ionized bubbles span sizes from a few cells to a sizeable fraction of the box, their boundaries are sharp on the grid scale, and whether two regions interact depends on the ionization state between them. This section reviews how existing architectures build in such structure, with the U-Net as the main example, and identifies what none of them provide.

2.1 Inductive biases: locality, scale and stationarity

Each layer of a neural operator can be written in the kernel form of Equation (1). Architectures differ mainly in which

kernels $\kappa(r, r', \theta)$ they allow, and these restrictions are their inductive biases. Three properties are useful for comparing them.

Receptive field. A convolutional neural network (CNN) uses kernels with compact support of a few grid cells, $\kappa(r - r') = 0$ for $|r - r'|$ larger than the stencil. A single layer is therefore strictly local; the receptive field grows only linearly with depth. The FNO is the opposite extreme: every Fourier layer couples every point to every other point.

Scale. A plain CNN or FNO processes the field at one resolution. Neither has a built-in notion of coarse and fine scales; scale-dependent behaviour has to be learned implicitly. Multi-scale architectures instead process the field at several resolutions and combine the results, so that large-scale context and small-scale detail are handled by separate parts of the network.

Stationarity and state dependence. CNNs, FNOs and U-Nets all use kernels that depend only on $r - r'$: the same filter is applied everywhere, and the coupling between two points is fixed once training ends. Attention mechanisms [9] and graph kernel networks [5] relax this by letting the kernel depend on the features at r and r', so that the coupling becomes *state-dependent*. The price is cost: a dense state-dependent kernel requires $\mathcal{O}(N^2)$ operations, which is prohibitive for three-dimensional grids with $N \sim 10^7$ cells.

Table 1 summarizes these properties for the architectures discussed in this report.

2.2 The U-Net

The U-Net [8] was introduced for biomedical image segmentation, a task that, like predicting an ionization field, requires sharp, pixel-accurate boundaries informed by large-scale context. It consists of three parts (Figure 2).

Contracting path. The encoder applies a stack of small convolutions followed by a nonlinearity, then halves the resolution by pooling, and repeats. Denoting the feature map at level s by e_s,

$$e_0 = C_0(a), \qquad e_s = C_s\big(\mathrm{Down}(e_{s-1})\big), \quad s = 1, \dots, S, \tag{6}$$

where each C_s is a block of 3×3 convolutions with nonlinearities. The number of channels is typically doubled at each level, so the network trades spatial resolution for feature richness. Because every level operates on a grid that is coarser by a factor of two, a 3×3 convolution at level s spans $3 \cdot 2^s$ cells of the original grid. The receptive field therefore grows exponentially with the number of levels instead of linearly with depth.

Bottleneck. At the coarsest level, e_S, the field is represented on a grid of $N/2^S$ points per dimension with many channels. Here a few convolutions see a large fraction of the domain and capture large-scale context.

Expanding path and skip connections. The decoder upsamples step by step and, at each level, concatenates the upsampled features with the encoder features of the same resolution:

$$d_S = e_S, \qquad d_s = C'_s\big(\big[\,\mathrm{Up}(d_{s+1}),\ e_s\,\big]\big), \quad s = S-1, \dots, 0, \tag{7}$$

and a final 1×1 convolution maps d_0 to the output u. The skip connections $e_s \to d_s$ are the key ingredient. Pooling discards the exact position of features, and upsampling alone cannot restore it. The skip connections pass the full-resolution encoder features directly to the decoder, so the network can combine *what* is present (from the coarse levels) with *where* it is (from the fine levels).

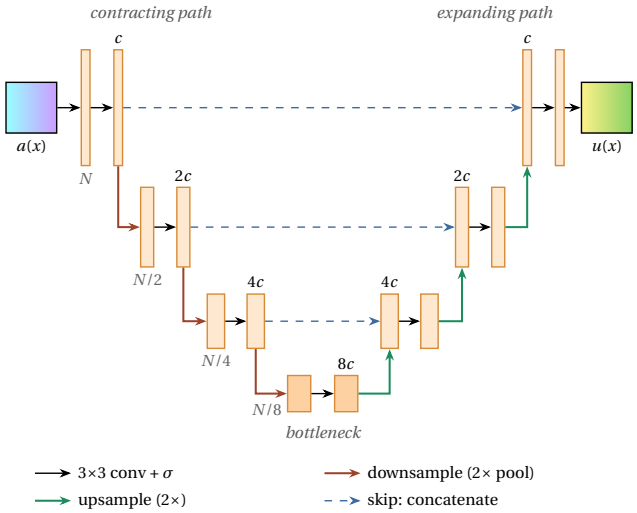


Figure 2. U-Net architecture with $S = 3$ downsampling steps. The contracting path (left) halves the resolution N and doubles the channel count c at each level; the expanding path (right) reverses this. Dashed skip connections concatenate encoder features with the upsampled decoder features of the same resolution, which restores the spatial detail lost by pooling.

Why it captures sharp structure. The U-Net addresses each limitation identified in Section 1.5. Its finest levels operate on the full grid with local kernels, so edges can be represented directly in physical space instead of through a truncated set of global modes. Its intermediate levels give it receptive fields at every scale between a few cells and the full box, which matches the broad range of bubble sizes. Finally, skip connections let high-frequency detail bypass the bottleneck entirely. U-Nets are therefore strong baselines for field-to-field emulation whenever sharp, localized features matter.

Limitations. The price is the loss of the properties that made the FNO attractive. The convolution kernels are defined in units of grid cells, not physical length, so changing the resolution changes the physical scale each filter responds to: the U-Net is not discretization invariant, and a model trained at one resolution cannot be reused at another. The number of pooling steps also fixes which grid sizes are admissible (multiples of 2^S). Global context is only available at the bottleneck and only at coarse resolution, so long-range interactions are approximated rather than represented exactly. And the U-Net is still stationary: the same filters are applied everywhere, independent of the local state of the field.

2.3 Hybrid architectures: U-FNO and related

The complementary strengths of the FNO (global, resolution-independent) and the U-Net (local, multi-scale) suggest combining them. The U-FNO of Wen et al. [10]

Table 1. Inductive biases of the operator architectures discussed in this report.

Architecture	Receptive field	Stationary	Multi-scale	State-dependent
CNN	local, grows with depth	yes	no	no
FNO	global (band-limited)	yes	no	no
U-Net	local → wide via pooling	yes	yes	no
U-FNO	global + multi-scale local	yes	yes	no
Attention	global	no	no	yes, $\mathcal{O}(N^2)$
Rhizome	global	no (gated)	implicit	yes, $\mathcal{O}(N \log N)$

does this inside the Fourier layer. In addition to the spectral path $\mathscr{F}^{-1}(R_\phi \cdot \mathscr{F} v)$ and the pointwise path $W v$ of Figure 1, each *U-Fourier layer* adds a small U-Net acting on v:

$$v_{\ell+1}(x) = \sigma\Big(W v_\ell(x) + \mathscr{F}^{-1}\big(R_\phi \cdot \mathscr{F} v_\ell\big)(x) + \mathrm{UNet}_\ell[v_\ell](x)\Big), \tag{8}$$

where UNet_ℓ denotes the U-Net path. The spectral path supplies global, low-k structure, while the U-Net path restores the high-k, localized detail that mode truncation removes. In the original application to multiphase flow, where saturation fronts are sharp, this substantially improved accuracy over the plain FNO [10].

The improvement comes at a cost that matters when comparing architectures. The U-Net path adds a large number of parameters, and a U-FNO is typically much larger than an FNO of the same width and depth. In our own comparison (Section 4), the U-FNO reference model has roughly two orders of magnitude more parameters than the matched models. Better accuracy can therefore not be attributed to the architecture alone. We treat the U-FNO as a dense reference baseline and make architectural claims only between parameter-matched models. The U-FNO also inherits the U-Net's resolution dependence through its convolutional path, and remains stationary.

Other hybrids follow the same pattern of adding a local or multi-scale branch next to a global spectral one. They differ in which branch is used and where it is inserted, but they share the structure of Equation (8): a sum of fixed, stationary operators.

2.4 Open problem: structure that adapts to the field

All architectures above, apart from attention, share one property: the communication pattern between points is fixed once training is done. Whether a kernel is local or global, single- or multi-scale, the coupling between r and r' depends only on their separation (and the level in the hierarchy), never on the field itself. Any dependence on the current state has to be built up indirectly, through pointwise nonlinearities stacked over many layers.

For reionization, this is at odds with the physics. Ionizing photons from a source travel until they are absorbed, so whether a source at r' affects a point r depends on the neutral fraction along the way: through an ionized region the coupling is long-range, through neutral gas it is short. The relevant interaction kernel is therefore state-dependent. Attention can represent this but scales as $\mathcal{O}(N^2)$; the stationary architectures scale well but cannot represent it directly.

An operator suited to this problem should combine four properties: (i) global reach, so that interactions across the whole box are possible; (ii) state-dependent coupling, so that the effective connections change as the field evolves; (iii) a cost close to $\mathcal{O}(N \log N)$, so that three-dimensional grids remain tractable; and (iv) a parameterization that is independent of the grid, so that the discretization invariance of the FNO is kept. The rhizome operator, introduced in Section 3, is designed around these four requirements.

3 RHIZOME NEURAL OPERATOR APPROACH

The name is borrowed from botany. A rhizome is a root system without a trunk or a hierarchy: any point can sprout a connection to any other, and which connections grow depends on local conditions rather than on a fixed plan. The rhizome neural operator applies this idea to operator learning. Every grid cell can exchange information with every other cell, as in a Fourier layer, but how strongly a cell sends and how strongly it listens is decided anew at every step from the current state of the field. The connections themselves therefore change as the prediction evolves, even though the parameters are fixed after training.

This section derives the operator from the kernel integrals of Section 1, then describes the architecture (Figure 3), how it is evaluated efficiently, and the variants used in the experiments of Section 4.

3.1 Design principles

The design follows the four requirements formulated at the end of Section 2.4:

1. **Global reach.** Interactions between cells are carried by a learned kernel $\kappa_\theta(r - r')$ on the whole periodic domain, evaluated in Fourier space as in the FNO.

2. **State-dependent coupling.** Two learned *gates* modulate the kernel at the sending and the receiving end. They are recomputed from the current hidden state at every update, so the effective coupling between two cells depends on the field at both of them.

3. $\mathcal{O}(N \log N)$ **cost.** The gates act channel-wise on each cell, so they factor out of the integral. The interaction is then a pointwise multiplication, one convolution evaluated with the FFT, and another pointwise multiplication: the same cost as a Fourier layer.

4. **Grid-independent parameters.** The model has no convolution stencils, no pooling and no parameters tied to particular cells. The kernel is parameterized by its Fourier coefficients on the normalized domain, and everything else acts pointwise.

Two further choices complete the design. First, there is exactly *one* mechanism for spatial communication: the gated

integral. There is no separate local or convolutional branch next to it, as in the U-FNO; lifting, the update proposal and decoding all act on single cells. Second, the same update block is applied T times with shared weights. The model is a recurrent dynamical system on the hidden state rather than a stack of distinct layers.

3.2 Mathematical background: from kernel integrals to the rhizome

The rhizome is best understood as a specific member of the family of kernel integral operators introduced in Section 1.2. This subsection derives it from the linear integral operator Equation (1) in three steps. First, the kernel is allowed to depend on the field it acts on. Second, that dependence is restricted to a separable form that keeps the FFT evaluation of the FNO. Third, the resulting operator is iterated. Throughout, D is the periodic domain normalized to unit volume, and $v : D \to \mathbb{R}^{d_v}$ is a hidden field with d_v channels.

A hierarchy of kernels. The most general kernel integral layer acting on v lets the kernel depend on position and on the field at both ends of each connection:

$$(\mathcal{K}[v]\, v)(r) = \int_D \kappa\big(r, r', v(r), v(r'); \theta\big)\, v(r')\, \mathrm{d}r'. \tag{9}$$

Different neural operators correspond to different restrictions of κ (Table 2). If κ depends on r and r' only, Equation (9) is the linear operator Equation (1): the Green's function picture of Section 1.2. If it depends only on $r - r'$, it is a convolution and becomes the Fourier layer Equation (3). If it depends on the field values, the operator becomes nonlinear in v, since v now enters both as the integrand and through the kernel. Self-attention [9] is the best-known example: there $\kappa \propto \exp\big(q(v(r)) \cdot k(v(r'))\big)$, normalized over r', and evaluating it on a grid costs $\mathcal{O}(N^2)$.

The rhizome sits between the Fourier layer and attention. It keeps a stationary kernel but multiplies it by two factors, one that depends on the field at the receiving point and one that depends on the field at the sending point:

$$\kappa^{\mathrm{eff}}(r, r') = \mathrm{diag}\big(A(v(r))\big)\, \kappa_\theta(r - r')\, \mathrm{diag}\big(B(v(r'))\big), \tag{10}$$

with $A, B : \mathbb{R}^{d_v} \to (0,2)^{d_v}$ the receiver and source gates. Setting $A = B = 1$ recovers the Fourier layer. Compared with attention, the dependence on the two endpoints is *separable*: it is a product of a function of $v(r)$ and a function of $v(r')$. This is also the idea behind linear attention [4], which replaces the softmax by a product of feature maps. The rhizome applies it channel by channel and multiplies it by a learned relative-position kernel $\kappa_\theta(r - r')$.

Operator form. Write M_a for the multiplication operator $(M_a f)(r) = a(r)\, f(r)$, applied channel by channel, and C_κ for the convolution $(C_\kappa f)(r) = \int_D \kappa(r - r') f(r')\, \mathrm{d}r'$. The rhizome message then reads

$$\boldsymbol{m} = M_{A(v)}\, C_{\kappa_\theta}\, M_{B(v)}\, V(v), \tag{11}$$

where V is the pointwise value map. Each factor is simple in one basis: the multiplication operators are diagonal in position space, and the convolution is diagonal in Fourier space, where $\widehat{C_\kappa f}(k) = \hat\kappa(k)\hat f(k)$. The rhizome therefore alternates between the two bases: multiply in position space, transform, multiply in Fourier space, transform back, multiply again. This is the same structure as the split-step Fourier methods of computational physics, where a potential acts in position space and the kinetic term in momentum space, each where it is diagonal. Like them, it costs two FFTs per application.

The factorization also makes the operator well behaved. Since the gates lie in $(0,2)$, both multiplication operators have norm at most 2 on $L^2(D)$. By Parseval's theorem, the norm of the convolution is $\sup_k \|\hat\kappa_\theta(k)\|$. Hence

$$\|M_A\, C_{\kappa_\theta}\, M_B\, f\|_{L^2} \;\le\; 4 \sup_{|k| \le k_{\max}} \|\hat\kappa_\theta(k)\|\, \|f\|_{L^2}, \tag{12}$$

independently of the field that produced the gates. State dependence cannot make the interaction blow up.

Mode coupling: smooth transport, sharp modulation. Section 1.4 noted that the spectral path of an FNO is band-limited: everything above $k_{\max}$ in its output has to be generated by the pointwise nonlinearity. In the rhizome, the gates change this. A product in position space is a convolution in Fourier space, so the Fourier transform of Equation (11) is

$$\begin{aligned} \hat{\boldsymbol{m}}(k) &= \sum_q \hat A(k - q)\, \hat\kappa_\theta(q)\, \widehat{(B\,V)}(q), \\ \hat\kappa_\theta(q) &= 0 \quad \text{for } |q| > k_{\max}. \end{aligned} \tag{13}$$

The kernel only transports the band $|q| \le k_{\max}$. Even so, the source gate first folds all input wavenumbers into that band, through $\widehat{BV}(q) = \sum_p \hat B(q - p)\hat V(p)$, and the receiver gate then spreads the transported band over every wavenumber present in A. If the receiver gate is sharp, for example close to zero inside an ionized region and close to two outside, a smooth message is multiplied by a sharp mask, and the result has a sharp edge exactly where the gate changes. The rhizome can thus carry information smoothly over long distances with few modes while placing sharp structure where the state of the field puts it. Whether the trained gates actually behave like this has not been checked directly; Equation (13) shows only that the mechanism is available.

Iteration as a path series. A single application of Equation (10) couples r' to r through the gates at the two endpoints only. Iteration adds paths. Ignoring for a moment the nonlinearity and the damping of the update, and freezing the gates, the recurrence becomes $v^{t+1} = c + \mathcal{K} v^t$ with $\mathcal{K} = M_A C_{\kappa_\theta} M_B$. After T steps this gives

$$v^T = \sum_{n=0}^{T-1} \mathcal{K}^n c \;+\; \mathcal{K}^T v^0, \tag{14}$$

the truncated Neumann series of the solution $v = (I - \mathcal{K})^{-1} c$ of the Fredholm integral equation of the second kind $v = c + \mathcal{K} v$. The resolvent $(I - \mathcal{K})^{-1}$ plays the role of the Green's function of Section 1.2, now for a state-dependent problem. Each term of the series is a sum over paths: the second-order term, for example,

$$\begin{aligned} (\mathcal{K}^2 f)(r) = \int_D\!\int_D A(r)\, \kappa_\theta(r - r_1)\, \underbrace{B(r_1) A(r_1)}_{\text{gates at } r_1} \\ \times\, \kappa_\theta(r_1 - r_2)\, B(r_2)\, f(r_2)\, \mathrm{d}r_1\, \mathrm{d}r_2, \end{aligned} \tag{15}$$

Table 2. Kernel integral operators ordered by what the kernel may depend on. N: number of grid points.

Operator	Kernel κ	Linear	Stationary	Cost
linear (Green's function)	$\kappa(r, r')$	yes	no	$\mathcal{O}(N^2)$
Fourier layer	$\kappa(r - r')$	yes	yes	$\mathcal{O}(N \log N)$
rhizome	$\operatorname{diag} A(v(r))\, \kappa_\theta(r - r')\, \operatorname{diag} B(v(r'))$	no	no	$\mathcal{O}(N \log N)$
attention	$\kappa(v(r), v(r'))$, normalized	no	no	$\mathcal{O}(N^2)$

couples r_2 to r through every intermediate point r_1, weighted by the gates there. This is the mathematical form of the path effects mentioned in Section 3.3. A region whose gates are small blocks the paths through it and so shadows what lies behind it, even though no single update can see beyond its two endpoints. In scattering theory the same expansion is the Born series, with each term adding one more scattering event. The actual rhizome differs from this linear picture through the tanh proposal, the damping $(1 - \Delta t)$ and the gates being recomputed at every step, but the path structure is the same.

Continuous-time view. The update of Algorithm 1, $\boldsymbol{h}^{t+1} = (1 - \Delta t)\boldsymbol{h}^t + \Delta t\, \boldsymbol{p}^t$, is the forward Euler discretization of the nonlocal evolution equation

$$\partial_\tau \boldsymbol{h}(r, \tau) = -\boldsymbol{h}(r, \tau) + \tanh\Big(\Phi\big(\boldsymbol{h}(r, \tau),\, (\mathscr{K}[\boldsymbol{h}]\, V(\boldsymbol{h}))(r),\, \boldsymbol{c}(r)\big)\Big), \tag{16}$$

integrated from $\tau = 0$ to $\tau = T\Delta t$ (3 in the experiments), with Φ the pointwise proposal network. Equations of this form, a decay term plus a saturating function of a spatially integrated input, are the neural field equations of computational neuroscience [1], in which $\partial_t u = -u + \int w(r - r')\, S(u(r'))\, \mathrm{d}r' + I(r)$ describes the activity of a continuous sheet of neurons. The rhizome generalizes that picture with many channels, a learned kernel and connections gated by the state. Viewed this way, it is also a neural ODE [3] on a function space, integrated with a fixed explicit scheme. The bound on the hidden state follows directly: if $|\boldsymbol{h}| \le 1$ componentwise, then so is $(1 - \Delta t)\boldsymbol{h} + \Delta t\, \boldsymbol{p}$ for $|\boldsymbol{p}| \le 1$ and $0 < \Delta t \le 1$.

Discretization. On a uniform grid of N points on the unit-volume domain, the integral in Equation (9) becomes the sum Equation (4) with weights $\Delta r_j = 1/N$. With the kernel represented by its retained Fourier modes, $\kappa_\theta(r) = \sum_{|k| \le k_{\max}} \hat{\kappa}_\theta(k)\, e^{2\pi i k \cdot r}$, this sum is evaluated *exactly* by the FFT:

$$\frac{1}{N} \sum_{j=1}^{N} \kappa_\theta(r_i - r_j)\, g(r_j) = \sum_{|k| \le k_{\max}} \hat{\kappa}_\theta(k)\, \hat{g}_k\, e^{2\pi i k \cdot r_i}, \qquad \hat{g}_k = \frac{1}{N} \sum_{j=1}^{N} g(r_j)\, e^{-2\pi i k \cdot r_j}, \tag{17}$$

provided every dimension of the grid has enough points (at least $2k_{\max}$ in the implementation) for the retained modes to remain distinct. The only approximation relative to the continuum operator is the quadrature itself. This identity is what the numerical verification in Section 3.6 checks, by evaluating the left-hand side directly over all pairs.

3.3 State-dependent integral interaction

Each grid cell i at position r_i carries a hidden state $\boldsymbol{h}_i^t$ of d_v channels and a fixed conditioning vector $\boldsymbol{c}_i$, built from the input density and the global parameters. At update t, cell i receives the message

$$\boldsymbol{m}_i^t = \frac{1}{N} \sum_{j=1}^{N} \operatorname{diag}\big(A_\theta(\boldsymbol{h}_i^t, \boldsymbol{c}_i)\big)\, \kappa_\theta(r_i - r_j) \times \operatorname{diag}\big(B_\theta(\boldsymbol{h}_j^t, \boldsymbol{c}_j)\big)\, V_\theta(\boldsymbol{h}_j^t, \boldsymbol{c}_j), \tag{18}$$

where

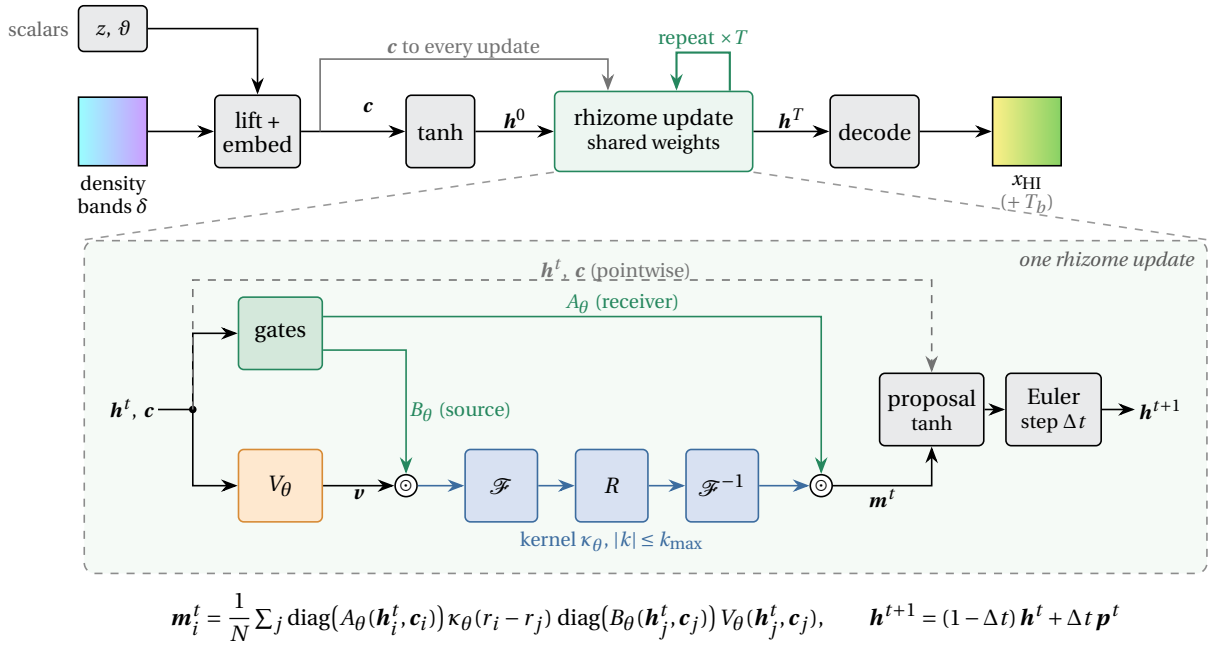
- V_θ builds the *value* that cell j transmits from its state and conditioning,
- B_θ is the *source gate*: how strongly cell j transmits, per channel,
- $\kappa_\theta(r_i - r_j) \in \mathbb{R}^{d_v \times d_v}$ is a learned, periodic, channel-mixing kernel that carries the signal across the separation $r_i - r_j$,
- A_θ is the *receiver gate*: how strongly cell i accepts what arrives, per channel,
- N is the number of grid cells, so that $1/N$ is the uniform quadrature weight on the normalized periodic box.

Read from right to left, Equation (18) describes a transmission: cell j prepares a value, decides how loudly to send it, the kernel carries it over the distance $r_i - r_j$, and cell i decides how much of it to take in. It is the quadrature of the operator Equation (11), with the effective kernel Equation (10) evaluated at the current state $(\boldsymbol{h}^t, \boldsymbol{c})$. This is the key difference from the FNO: the base kernel κ_θ is stationary, but the effective kernel depends on the field at both ends of every connection and changes from update to update.

The gates are deliberately simple. They act channel by channel on single cells and therefore factor out of the sum in Equation (18). This is what keeps the cost at that of a Fourier layer: the network never forms the $N \times N$ matrix of pairwise couplings, which for a 140×140 slice alone would have about 3.8×10^8 entries per channel pair. The price is expressivity. The coupling between two cells is a product of a sender factor and a receiver factor around a fixed kernel, not an arbitrary function of both states as in attention. In particular, a single update cannot make the coupling between r and r' depend on the cells *between* them. Such path effects, like the shadowing of a region by neutral gas along the line of sight, emerge only over several updates, through the path series Equation (15).

3.4 Gating: source and receiver factors

A single pointwise network maps $[\boldsymbol{h}^t, \boldsymbol{c}]$ to $2d_v$ logits per cell, which are split into the source and receiver gates and passed through $2\,\mathrm{sigmoid}(\cdot)$. Each gate value therefore lies in $(0, 2)$ and equals one at zero logit: an untrained gate leaves the



$$m_i^t = \frac{1}{N}\sum_j \mathrm{diag}\big(A_\theta(h_i^t, c_i)\big)\,\kappa_\theta(r_i - r_j)\,\mathrm{diag}\big(B_\theta(h_j^t, c_j)\big)\,V_\theta(h_j^t, c_j), \qquad h^{t+1} = (1-\Delta t)\,h^t + \Delta t\,p^t$$

Figure 3. Rhizome Neural Operator architecture. *Top:* the input density bands and the global scalars (redshift z, simulation parameters ϑ) are lifted and embedded into a conditioning field $\boldsymbol{c}$, which initializes the hidden state and is fed into every update. One update block with shared weights is applied T times; a pointwise decoder maps the final state to x_{HI} (and, in the multi-field model, to T_b). *Bottom:* one update. The gates produce a source factor B_θ and a receiver factor A_θ from the current state. The source-gated value is convolved with the kernel κ_θ in Fourier space and multiplied by the receiver gate, giving the message $\boldsymbol{m}^t$. A pointwise proposal combines message, state and conditioning, and an explicit Euler step updates the state.

kernel unchanged, and training can then learn to amplify or suppress transmission and reception.

Three properties of the gates are worth stating explicitly:

- **They are recomputed at every update** from the previous state and the fixed conditioning. A cell that becomes ionized during the rollout can change how it sends and receives in the next update.
- **Source and receiver are independent.** Interactions may be asymmetric: cell j can influence cell i strongly while i barely influences j. Since the kernel is not constrained to be positive, interactions may also be signed.
- **They are channel-wise.** Each of the d_v channels has its own gate, so different channels can carry different kinds of information with different reach. Channels are mixed only by the kernel κ_θ and by the pointwise networks.

No physical constraint is imposed: the operator does not enforce reciprocal couplings, conservation of photons or any form of radiative transfer. Whatever physical interpretation the gates acquire is learned from the data.

3.5 Recurrent time-stepping

The hidden state is initialized from the conditioning and updated T times with the same weights (Algorithm 1). Each update forms the message, combines it with the state and the conditioning in a pointwise *proposal*, and takes an explicit Euler step of size Δt towards that proposal. The final state is decoded pointwise into the x_{HI} logit.

Because the proposal is bounded by tanh and the update is a convex combination for $0 < \Delta t \le 1$, the hidden state stays bounded by one throughout the rollout. This keeps long rollouts numerically stable, but it is not a convergence guarantee: nothing forces the iteration towards a fixed point. In practice the model is trained and evaluated at a fixed number of updates. All experiments in Section 4 use $T = 6$ and $\Delta t = 0.5$.

Algorithm 1 Rhizome operator forward pass

Require: density bands δ, redshift z, parameters ϑ
$\boldsymbol{c} \leftarrow \mathrm{lift}(\delta) + \mathrm{embed}(z, \vartheta)$
$\boldsymbol{h}^0 \leftarrow \tanh(\boldsymbol{c})$
for $t = 0, \dots, T-1$ **do** ▷ shared weights
  $(\boldsymbol{s}, \boldsymbol{r}) \leftarrow 2\,\mathrm{sigmoid}\big(\mathrm{gates}([\boldsymbol{h}^t, \boldsymbol{c}])\big)$ ▷ source, receiver
  $\boldsymbol{v} \leftarrow V_\theta([\boldsymbol{h}^t, \boldsymbol{c}])$
  $\boldsymbol{m}^t \leftarrow \boldsymbol{r} \odot \mathscr{F}^{-1}\big(R \cdot \mathscr{F}(\boldsymbol{s} \odot \boldsymbol{v})\big)$
  $\boldsymbol{p} \leftarrow \tanh\big(\mathrm{pointwise}([\boldsymbol{h}^t, \boldsymbol{m}^t, \boldsymbol{c}])\big)$
  $\boldsymbol{h}^{t+1} \leftarrow (1-\Delta t)\,\boldsymbol{h}^t + \Delta t\,\boldsymbol{p}$
end for
return $x_{\mathrm{HI}} = \mathrm{sigmoid}\big(\mathrm{decode}(\boldsymbol{h}^T)\big)$

Weight sharing makes the parameter count independent of T. It also makes the model behave like a learned dynamical system rather than a deep feed-forward network, which raises the question of what happens away from the training horizon. A small pilot on synthetic 32×32 lightcones answers this (Table 3). The tied rhizome improved modestly on an earlier additive recurrent model (Fourier message plus separate local message) at an equal parameter count, while an untied version with independent weights per update, at four times the parameters, was more accurate still. Running the tied model for more updates than it was trained with degraded its accuracy, although the hidden state stayed bounded. The number of updates is therefore

part of the trained model, not a free accuracy knob.

Table 3. Architecture pilot on synthetic 32 × 32 lightcones (single seed, CPU; 512 training slices, 64 test slices). *Top:* rhizome against the additive recurrent baseline. *Bottom:* the tied rhizome evaluated at different numbers of updates; it was trained with four. Mixed slices have mean $x_{\rm HI}$ between 0.05 and 0.95. This toy problem is not a physical simulation; the table characterizes the architecture, not its accuracy on 21 cm data.

Model	Params	RMSE, mixed
Rhizome, tied	15.6 k	0.196
Additive recurrent	15.9 k	0.207
Rhizome, untied	60.3 k	0.181

Updates T	RMSE, mixed
1	0.260
2	0.205
4 (trained)	**0.196**
8	0.265

3.6 Discretization and spectral evaluation

The FFT in Figure 3 is not a separate component that predicts anything; it is simply an efficient and exact way to evaluate the sum over j in Equation (18) (Equation (17)). The kernel κ_θ is parameterized by its Fourier coefficients $R(k)$ for $|k| \le k_{\max}$, exactly as in the FNO (Equation (5)). Three details make this evaluation correct:

- **Quadrature.** With an orthonormal FFT pair, the forward and inverse transforms together already produce the $1/N$ weight of Equation (18). Adding a further factor for the cell area would make messages shrink as the resolution grows.
- **Real kernel.** The zero mode and the conjugate-symmetric modes are constrained so that κ_θ is real in physical space.
- **Verification.** The FFT implementation was checked against an independent calculation of the left-hand side of Equation (17) that sums the retained Fourier series over all source–receiver pairs explicitly, without any FFT. Outputs agree to about 10^{-15}, and so do the gradients with respect to inputs and parameters.

The architecture inherits the FNO's discretization properties (Section 1.4). Since no parameter refers to a particular cell, shifting the input periodically shifts the output in the same way. The same weights can be applied to any grid that represents the same physical box with at least $2k_{\max}$ points per dimension. The caveats also carry over: the gates and the proposal are nonlinear and alias on coarse grids, so transfer across resolutions has to be validated rather than assumed. It has not been tested in this work.

3.7 Model variants

The same operator was used in four configurations, summarized in Table 4. They differ in what the model sees along the line of sight (LOS) and in which fields it predicts.

2-D slices with LOS context. The first variant predicts one transverse 140 × 140 slice of a lightcone. Its input is the density on that slice and on the neighbouring slices: single slices within ±3 cells and band means further out, up to ±32 cells, for 13 density channels in total. The conditioning scalars are $\log(1+z)$ and the eleven simulation parameters.

3-D native windows. The second variant works directly in three dimensions on windows of the native lightcone. It uses anisotropic mode counts (36 in each transverse direction, 24 along the LOS) and a rank-48 factorized spectral kernel to keep the parameter count at 5.8 M. Because the LOS is not periodic, the windows are zero-padded by 64 cells before the FFT. A code review found that this padding is too short to separate all physical LOS offsets: displacements of +160 and −160 cells fall on the same residue of the padded circle, so the base kernel cannot treat them differently. The gates can still modulate these couplings, but the 3-D kernel is a constrained rather than a fully general LOS interaction.

Slice-wise reconstruction of 3-D cones. The third variant applies the 2-D slice model to every native slice of a lightcone, each with its own ±32-cell LOS context, and restacks the predictions into a full 3-D cone (Figure 4). In the language of windowed inference this is maximal overlap: window one slice, stride one slice. The model never sees the full LOS jointly; it only needs to know what lies within about 46 Mpc of each slice. As Section 4 shows, this turns out to be enough, and it avoids both the padding issue above and the memory cost of 3-D windows.

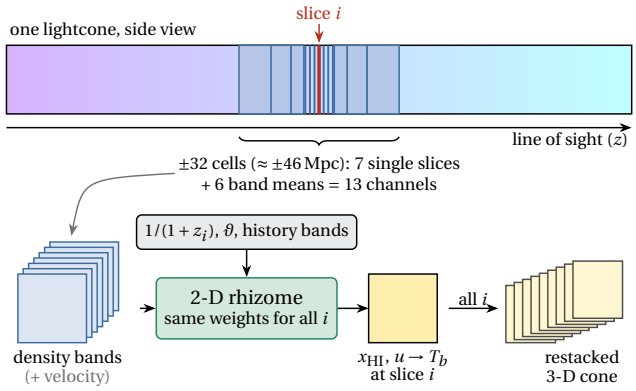


Figure 4. Slice-wise reconstruction of a 3-D lightcone. For each slice i along the line of sight, the 2-D rhizome receives 13 density channels: the seven single slices within ±3 cells and six band means out to ±32 cells (blue). In the multi-field model the same bands are built for the LOS velocity. Global scalars enter as conditioning. The prediction for slice i is computed independently, and the predictions for all slices are restacked into the 3-D cone.

Multi-field prediction. The final variant predicts both $x_{\rm HI}$ and the 21 cm brightness temperature T_b. It receives 13 LOS bands each of density and LOS velocity. Its scalars are $1/(1+z)$, the eleven parameters and the band means of two emulated global histories, $\bar{x}_{\rm HI}(z)$ and $\bar{T}_b(z)$. The model outputs an $x_{\rm HI}$ logit and a spin variable u, and T_b is assembled by a *structured head* that encodes the known form of the signal:

$$T_b = T_0(z)\, x_{\rm HI}\, (1+\delta)\, F_\nu \left(1 - e^{u}\right), \tag{19}$$

Table 4. Rhizome variants used in the experiments. LOS: line of sight.

Variant	Input	Output	Kernel
2-D slice	one transverse slice + 13 LOS density bands (±32 cells)	x_{HI} on the slice	2-D, 24 modes
3-D window	native LOS windows of 256 cells + 2×32 halo	x_{HI} in the window	3-D, modes $(36, 36, 24)$, rank-48 factorized
Slice-wise 2-D	2-D slice model applied to every slice of a cone	full 3-D x_{HI} cone, restacked	as 2-D slice
Multi-field slice-wise	density and LOS-velocity bands, global-history band means	x_{HI} and T_b (structured head)	as 2-D slice

where $T_0(z)$ is the redshift-dependent amplitude, F_ν the correction for the LOS velocity gradient, and $1 - e^u$ plays the role of the spin-temperature factor $1 - T_{\mathrm{CMB}}/T_S$. The physical prefactor $T_0(1+\delta)F_\nu$ is computed exactly, so the network only has to learn the neutral fraction and the spin factor.

3.8 Assumptions and limitations

The rhizome makes fewer structural assumptions than the architectures of Section 2, but it does make some:

- **Periodicity.** The kernel lives on a periodic domain. This is exact in the transverse directions of the lightcones, which are periodic, but not along the LOS. The 2-D and slice-wise variants avoid the issue by treating the LOS through input bands; the 3-D variant relies on zero padding (Section 3.7).
- **Product-form coupling.** The effective kernel Equation (10) is a stationary kernel modulated at its two ends. It cannot represent couplings that depend jointly and arbitrarily on the states of both cells, or on the cells between them, within a single update.
- **No physics in the operator.** Reciprocity, conservation and radiative transfer are not imposed. Physics enters only through the inputs, the conditioning and, for T_b, the structured head.
- **Fixed rollout length.** The model is accurate at the number of updates it was trained with, not beyond (Table 3). There is no convergence or fixed-point guarantee.
- **Untested resolution transfer.** Although the parameters are grid-independent, all results in this report were obtained at a single resolution.

3.9 Relation to FNO, U-FNO and attention

Table 1 places the rhizome among the architectures of Section 2. Compared with the **FNO**, it keeps the global spectral kernel and its $\mathcal{O}(N \log N)$ cost, adds the gates that make the effective kernel state-dependent, and replaces the stack of independent layers by a weight-shared recurrence. The pointwise proposal takes over the role of the FNO's pointwise path W. With gates fixed to one and untied weights, an update reduces to a Fourier layer with a nonlinear pointwise path.

Compared with the **U-Net and U-FNO**, it has no convolution stencils, no pooling and no explicit hierarchy of scales. Multi-scale behaviour has to come from the kernel, which spans all scales up to $k_{\max}$, and from the repeated updates. This keeps the parameterization independent of the grid, which the convolutional path of the U-FNO does not.

Compared with **attention**, it shares the central idea that the coupling between two points should depend on their content, but it restricts that dependence to a product of per-cell factors around a stationary kernel. This restriction is what brings the cost down from $\mathcal{O}(N^2)$ to $\mathcal{O}(N \log N)$: the rhizome trades the full flexibility of pairwise attention for a form that can be evaluated with a single FFT pair per update.

The cost of a forward pass is T times that of one Fourier layer, $\mathcal{O}(T\, d_v\, N \log N)$ for the transforms plus pointwise work, and the parameter count is that of a single update block.

4 TRAINING AND RESULTS

This section reports how the rhizome was trained and how it compares with the best models of the existing `fno-21cm` pipeline on the task of predicting the neutral-fraction field x_{HI}, and later also the brightness temperature T_b, from the matter density of `21cmFAST` lightcones. The experiments followed the order of Table 4: 2-D slices, native 3-D windows, multi-field 3-D, and finally slice-wise reconstruction of full cones. Figure 5 summarizes the outcome.

4.1 Datasets

All data come from one suite of 6,600 `21cmFAST` [7] simulations (version 4.1.1), each with eleven sampled simulation parameters. Every simulation provides a 3-D lightcone of 140×140 transverse cells over a 200 Mpc box (1.43 Mpc per cell). The native line-of-sight length depends on cosmology and is between 2,074 and 2,919 cells at the same spacing. The lightcones contain the density, the neutral fraction, the brightness temperature and the LOS velocity, among other fields. Two properties of the data matter for the models:

- Each lightcone repeats one periodic 200 Mpc box along the LOS, so any 140 consecutive LOS cells contain a full, cyclically shifted copy of the box.
- Ionized cells have x_{HI} exactly zero; partially ionized values appear mostly where the lightcone interpolates between two simulation snapshots.

33 simulations (0.5%) contain non-finite T_b values and are excluded from every multi-field split. Table 5 lists the splits used.

The x_{HI} and multi-field splits are those of the `fno-21cm` pipeline. All comparisons with `fno` models below use identical test cones and the same evaluation code.

Table 5. Training, validation and test splits. All test sets are held out by simulation (cone), never by slice.

Split	Used for	Train / val / test cones	Samples
2-D slice cache	2-D sweep and first 2-D model	5,280 train (own split)	16 window + 2 background slices per cone
x_{HI} (fno split)	3-D rhizome; fair slice-wise model	1,600 / 200 / 200	3-D windows; or 48 + 6 slices per cone
Multi-field	3-D multi-field; slice-wise multi-field	1,594 / 199 / 199	as above, cleaned T_b
Multi-field, all cones	best slice-wise multi-field model	6,169 / 199 / 199	16 + 2 slices per cone (98.5 k)

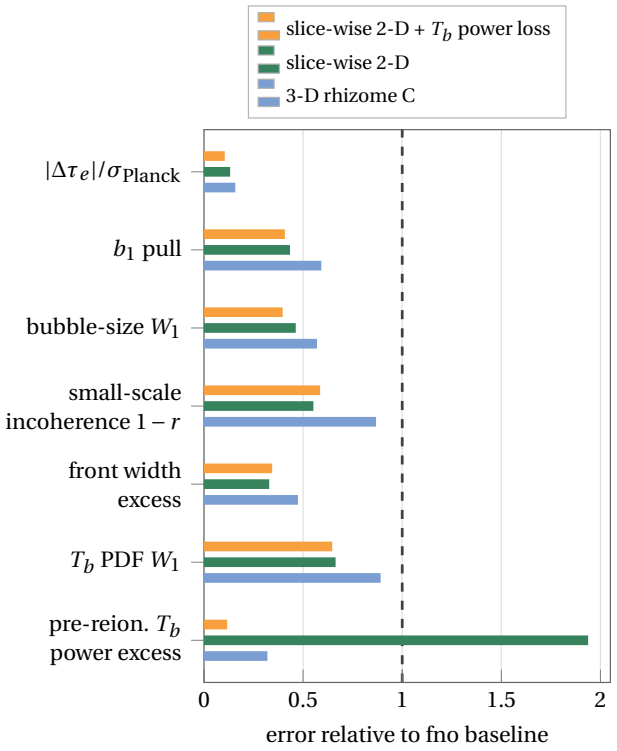


Figure 5. Headline results on 199 held-out multi-field test cones. Each bar is the error of a rhizome model divided by the error of the best `fno-21cm` multi-field model (`tbs+histtb`); values below the dashed line are better than the baseline. Front width is measured as the excess over the true front width of 2.77 Mpc. The slice-wise model wins every metric except the small-scale T_b power before reionization, which a power term in the loss repairs (Section 4.10). Numbers in Table 12.

4.2 Training protocol

Losses. x_{HI} is trained with binary cross-entropy (BCE) on the logit. In the multi-field models, T_b is added as a mean-squared error on the normalized field, through the structured head of Equation (19). The choice of BCE turned out to matter beyond the operator (Section 4.7). The `fno` baselines train x_{HI} with a mean-squared error through a sigmoid, whose gradient vanishes when the prediction saturates.

Optimization and stopping. The learning rate is 2×10^{-3} throughout; the 3-D models use warmup and cosine decay. The 2-D models are stopped by a predefined plateau rule on the validation loss, and the checkpoint with the best validation score is kept. The 3-D models are trained for 20 epochs of 6,400 windows and then continued with a short *warm restart* (10 epochs at 5×10^{-4} with cosine decay), which improved test RMSE by 3–6%, against about 1% for a continuation at constant small learning rate. The 2-D models use the dihedral symmetries of the square slice as data augmentation.

Hyperparameters. Table 6 lists the main settings. Every model is a single training run with one seed.

4.3 Evaluation metrics

Per-voxel RMSE is reported, but it is a weak guide to physical quality: a model can lower its RMSE by blurring fronts. The evaluation therefore uses the physical suite of the `fno-21cm` pipeline, computed on full test cones. Most metrics are evaluated in the *active* stage of reionization, slices with mean neutral fraction between 0.05 and 0.95, and some also *before reionization*, $\bar{x}_{\mathrm{HI}} \geq 0.98$, where T_b is in absorption. Table 7 lists them.

The cross-correlation coefficient between prediction and truth,

$$r(k) = \frac{P_{\mathrm{pred}\times\mathrm{true}}(k)}{\sqrt{P_{\mathrm{pred}}(k)\,P_{\mathrm{true}}(k)}}, \tag{20}$$

deserves special attention. The power ratio measures whether a model produces the right *amount* of structure at each scale; $r(k)$ measures whether that structure is in the right *place*. In the earlier 3-D operator study of the `fno-21cm` pipeline, small-scale coherence predicted the bias in the bubble-size distribution across architectures (Spearman $\rho = 0.88$), while small-scale power amplitude did not. Getting fronts in the right place matters more for bubble statistics than getting the right amount of small-scale detail.

4.4 Baselines

All baselines come from the `fno-21cm` pipeline and were trained there on the same splits:

- `cnn_whno_es` (3.5 M parameters): the `fno` 3-D x_{HI} model trained on native LOS windows. It has the U-FNO layout, a convolutional local path next to a learned global path, at a size comparable to the rhizome. Trained for 30 epochs (44.7 h on an A100).
- `es+histtb` and `tbs+histtb`: the corresponding multi-field models, which receive density, velocity and both emulated global histories. `tbs+histtb` additionally uses the structured T_b head and is the strongest `fno` model. It corresponds to rhizome variant C below and is labelled "UFNO" in the head-to-head figures.
- **2-D `fno` models** (LWF, WHNO, LocalFNO, U-FNO), which predict one slice from that slice alone.

Table 6. Main hyperparameters.

	2-D slice / slice-wise models	3-D window model
width d_v	64 (128 in early runs)	48
Fourier modes $k_{\max}$	24	(36, 36, 24), rank-48 factorized
updates T / step Δt	6 / 0.5	6 / 0.5
batch / sample	32 slices	native window, $140\times140\times256 + 2\times32$ halo
learning rate	2×10^{-3}, plateau-stopped	2×10^{-3} (warmup + cosine), 20 epochs + warm restart
precision	float32	bfloat16 autocast, float32 FFT
parameters	grow as d_v^2: 1.1 M (w32) to 18.3 M (w128)	5.8 M
training time	2.8–3.5 h (A100)	~12 h + 6 h restart (H200)

Table 7. Evaluation metrics. All are computed per test cone and summarized by the median over cones.

Metric	What it measures
$\lvert\Delta\tau_e\rvert/\sigma_{\text{Planck}}$	error in the CMB optical depth implied by the predicted global history, in units of the Planck uncertainty
$\lvert\Delta z\rvert$ at $\bar{x}$ = 0.25/0.5/0.75; duration	timing of reionization: error in the redshift at which the global neutral fraction crosses each value, and in the duration
x_{HI} PDF W_1	Wasserstein-1 distance between predicted and true one-point distributions of x_{HI}
partially ionized fraction	fraction of cells with intermediate x_{HI} (truth ≈ 0.09); measures hedging
b_1 pull; P_ε ratio	EFT diagnostics: error in the fitted linear bias in units of its uncertainty; ratio of stochastic power to the truth (ideal 1)
bubble-size W_1	Wasserstein-1 distance (Mpc) between mean-free-path bubble-size distributions
power error (large / mid / small k)	$\lvert P_{\text{pred}}/P_{\text{true}} - 1\rvert$ of the transverse x_{HI} power spectrum in three k ranges; small means $k > 1\,\text{Mpc}^{-1}$
$1-r$, small k	small-scale incoherence: one minus the cross-correlation Equation (20), averaged over $k > 1\,\text{Mpc}^{-1}$
edge L^2 / H^1, ±2 Mpc	error (and error in gradients) in a ±2 Mpc band around true ionization fronts
front width	width of the predicted ionization fronts (Mpc), compared with the true width
T_b PDF W_1; T_b power deviation	distance between T_b distributions (mK); power difference in units of the single-slice sample variance (σ)

The 203 M-parameter U-FNO of Wen et al. [10] from the operator study (Section 2.3) is not used as a baseline here, because its capacity is not comparable.

4.5 2-D slices: line-of-sight context dominates

A sweep around a centre configuration (width 64, 24 modes, six updates) showed that the largest single factor is not a hyperparameter of the operator but the LOS context in the input (Figure 6). With only the target slice, the rhizome reaches a mixed-slice RMSE of 0.174. Adding the six neighbouring slices brings it to 0.138, and adding the band means out to ±32 cells to 0.112, a 36% reduction in total. By comparison, the other settings matter little: width 128 is 3% better than width 64 at four times the parameters, 12 or 48 modes are worse than 24, and three or ten updates are slightly worse than six. At equal compute, more training slices beat more width.

Table 8 compares the 2-D rhizome with the 2-D `fno` models on the same 7,036 validation slices. The rhizome halves the error of the next best model, and its ionization fronts are five times sharper. Most of this gap, however, is LOS information: the `fno` models see a single slice, and the single-slice rhizome (0.174) performs at their level. The

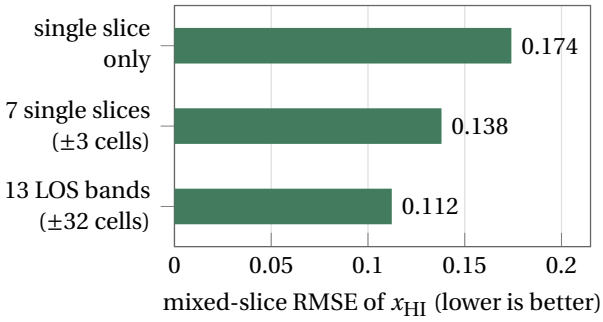


Figure 6. Effect of line-of-sight context on the 2-D rhizome (sweep, 15 k steps, one seed). Mixed slices have mean x_{HI} between 0.05 and 0.95.

2-D comparison therefore shows what the input buys rather than what the operator buys. The fully trained 2-D model (width 128, plateau-stopped at 68 k steps) reached a test mixed-slice RMSE of 0.099 and an ionized-region IoU of 0.914.

4.6 3-D windows: x_{HI}

The 3-D rhizome sees the same information as the 3-D `fno` model: native LOS windows of the density field. Here the

Table 8. 2-D rhizome against 2-D `fno` models on 7,036 validation slices (rhizome: sweep run at 15 k steps). Only the rhizome sees LOS context. RMSE: slice RMSE; IoU: ionized-region intersection over union; front: front width (truth 3.0 Mpc).

Model	RMSE	IoU	Front (Mpc)
Rhizome (LOS bands)	**0.109**	**0.873**	**6.1**
`fno` LWF	0.178	0.755	32.6
`fno` WHNO	0.181	0.755	31.0
`fno` LocalFNO	0.184	0.742	–
`fno` U-FNO	0.195	0.722	–

comparison isolates the operator. On the 200 test cones (Table 9), the rhizome at width 48 reaches an RMSE of 0.0594 after the warm restart, against 0.0684 for `cnn_whno_es`, with 5.8 M against 3.5 M parameters. Narrower was better (width 64 gave 0.0628), and a learning rate of 3×10^{-4} was 13% worse than 2×10^{-3}.

The margin is much larger on the physical metrics than on RMSE. The rhizome is three times more accurate in τ_e and in the reionization midpoint, halves the bias error, and comes 40% closer in bubble sizes. Its small-scale power and coherence also hold up better at every stage of reionization.

4.7 3-D multi-field: inputs, heads and losses

The multi-field models predict $x_{\rm HI}$ and T_b from density and LOS velocity. Four 3-D rhizome configurations were trained to separate the contributions of inputs and output head (Table 10).

- **History channels carry the gain.** Going from A to B reduces the $x_{\rm HI}$ error by 21% and the T_b error by 34%.
- **Velocity and a T_b target alone do not help** $x_{\rm HI}$; A is even slightly worse than A0 (one seed each).
- **The structured head mainly helps** T_b: without it, T_b power is about 10% low at every stage. It also improves the reionization timing by 15–30%.
- **The density-only rhizome A0 already beats** `fno es+histtb` (0.0567 vs 0.0577), although the latter also receives velocity and both histories.

On the physical suite, model C beats `tbs+histtb` by a factor of 5 in τ_e (0.014 vs 0.076 σ), and is closer in b_1 (pull 1.52 vs 2.28), P_ε (0.935 vs 0.793), bubble sizes (W_1 0.71 vs 1.21 Mpc) and front width (4.53 vs 6.17 Mpc; truth 2.77). Before reionization, `fno` adds spurious small-scale T_b structure (+12.8σ against +3.9σ).

Part of the gap is the loss, not the operator. Both model families receive the same emulated global history, yet the global $\bar{x}_{\rm HI}(z)$ of the `fno` models sits 0.002–0.003 too low from $z \approx 8$ to 25. In truly neutral cells at $z > 15$ the mean error is -0.0020 and -0.0030 for the two `fno` runs against -0.00006 for the rhizome. The cause is the `fno` loss: a mean-squared error through a sigmoid has a vanishing gradient near saturation, so fully neutral cells are left at about 0.997, while the gradient of BCE in logit space does not vanish. This uniform offset integrates into a τ_e bias. A fair comparison of τ_e and the global history therefore needs the `fno` models retrained with BCE. The timing, bias, stochasticity, bubble-size and front metrics, however, are not explained by this offset.

4.8 Slice-wise reconstruction beats native 3-D

The slice-wise variant (Figure 4) applies the 2-D model to every slice of a cone. For a fair comparison it was retrained on a new slice cache built from the same 1,600 training cones as the 3-D models, with otherwise unchanged settings. Table 11 compares it with the 3-D rhizome and with `fno` on all 200 test cones.

The slice-wise model wins on nearly every physical metric, trains about seven times faster than the 3-D rhizome, and has no jitter between slices: the roughness from slice to slice matches the truth to 2–3%, while both 3-D models are slightly over-smoothed along the LOS. Its advantage comes from front *placement*: small-scale incoherence drops from 0.20 to 0.12, the mid-scale power error by 60%, and the edge error by 21% (significant in a paired bootstrap over 175 cones). Width barely matters, so width 64 (2.8 h of training against 5.0 h) is the choice. Per cone, the 2-D model wins 11 of 13 inspected cones. It loses on the earliest reionizer, cone 27 ($z_{\rm mid} \approx 15.9$), where the longer context of the 3-D model helps slightly.

The error maps in Figure 7 show the qualitative difference. The slice-wise errors are thin rims along bubble walls, whereas the 3-D models, `fno` strongly and the 3-D rhizome weakly, miss whole ionized regions, and `fno` leaves a faint ionized haze over neutral gas.

This result says something about the problem as well as about the model. What the 3-D models need from the line of sight is about ±32 cells (≈ ±46 Mpc) of context, not a joint prediction over 256 cells. A 2-D operator with that context in its input is both more accurate and cheaper than a 3-D operator that has to learn LOS structure across a whole window.

4.9 Best model: multi-field slice-wise rhizome

The final model combines the slice-wise approach with the multi-field inputs and the structured head of model C: 13 LOS bands each of density and velocity, global-history band means as scalars, and $x_{\rm HI}$ and T_b as outputs (Section 3.7). It was trained on all 6,169 usable training cones (3.5 h on an A100) and evaluated on the 199 multi-field test cones. Table 12 compares it with 3-D rhizome C and with the best `fno` model.

Against `fno`, the slice-wise model reduces the $x_{\rm HI}$ value-distribution error by 65%, the b_1 pull by 57%, the bubble-size error by 54% and small-scale incoherence by 45%, and its fronts are 3.9 Mpc wide against 6.2 Mpc. During reionization, its T_b distribution is 34% closer to the truth. On the eight test cones inspected individually, it beats `fno` on every cone, by 12–28% in full-cone RMSE. It also beats the 3-D rhizome C on every $x_{\rm HI}$ morphology metric (value distribution −39%, b_1 pull −27%, bubble sizes −19%, incoherence −36%, edges −19%), while τ_e and timing tie: with history channels, both rhizome models are already five to seven times better than `fno` there. Training on all cones instead of the 1,594-cone split improved per-cone RMSE by 3.6% ($x_{\rm HI}$) and 5.5% (T_b) in the median, on 86% and 75% of cones respectively.

Figure 9 shows the scale dependence. Both models lose small-scale power late in reionization, `fno` more so. The rhizome keeps $r > 0.9$ to higher k at every stage, so the

Table 9. 3-D $x_{\rm HI}$ on 200 test cones. *Left:* test RMSE. *Right:* physical suite (medians over cones), rhizome w48 after the warm restart.

Model	Params	RMSE
rhizome w48, 20 ep.	5.8 M	0.0619
+ warm restart	5.8 M	**0.0594**
rhizome w64, + restart	–	0.0628
`cnn_whno_es`, 30 ep.	3.5 M	0.0684

Metric	Rhizome	`fno`
$\lvert\Delta\tau_e\rvert/\sigma_{\rm Planck}$	**0.019**	0.058
$\lvert\Delta z(\bar{x}=0.5)\rvert$	**0.028**	0.098
b_1 pull	**1.58**	3.41
P_ε ratio (ideal 1)	**0.89**	0.77
bubble-size W_1 (Mpc)	**0.81**	1.39
small-scale power error	**0.222**	0.336
edge H^1, ±2 Mpc	**0.137**	0.145
front width (Mpc)	**5.9**	7.5

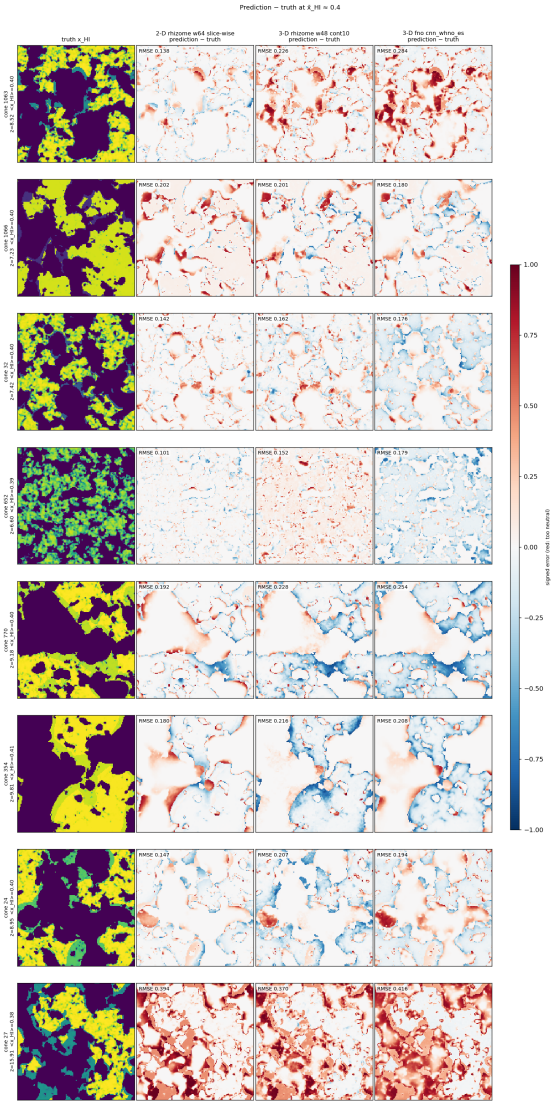


Figure 7. Signed $x_{\rm HI}$ error at $\bar{x}_{\rm HI}\approx0.4$ on eight test cones (red: predicted too neutral, blue: too ionized). Columns: truth, slice-wise 2-D rhizome (w64), 3-D rhizome (w48, warm restart), `fno` `cnn_whno_es`; per-panel RMSE in the corner. The bottom row is cone 27, the earliest reionizer, where all models predict ionized gas as too neutral.

Table 10. 3-D multi-field ablation, test RMSE on 199 cones ($x_{\rm HI}$ / T_b in mK). History channels are the band means of the emulated global $x_{\rm HI}(z)$ and $T_b(z)$.

Run	Inputs / head	20 epochs	+ restart
A0	density → $x_{\rm HI}$ only	0.0602	0.0567
A	+ velocity, + T_b (plain head)	0.0648 / 9.49	0.0622 / 8.71
B	A + history channels	0.0509 / 6.26	0.0494 / 6.07
C	B + structured T_b head	0.0507 / 5.97	**0.0495 / 5.79**
`fno es+histtb`	= B	0.0577 / 6.63	
`fno tbs+histtb`	= C	0.0560 / 6.54	

structure it does produce sits in the right place. Figure 10 puts this in the context of all models from the earlier 3-D operator study. In the plane of small-scale coherence against small-scale power error, the multi-field slice-wise rhizome sits apart from every `fno` model: $r(k>1)=0.90$ against 0.82–0.83, power error 0.17 against 0.31–0.34.

Figure 12 shows the signed error at $\bar{x}_{\rm HI}\approx0.4$ on eight test cones. As for the $x_{\rm HI}$-only model, the rhizome's errors are thin rims on bubble walls, while `fno` misses whole regions and its rims are wider.

Shared failure. Some test cones contain large, flat plateaus of intermediate neutral fraction ($x_{\rm HI}\approx0.3$–0.6) in the truth. No model reproduces them, and all overpredict the partially ionized fraction (0.13–0.18 against 0.09).

4.10 Brightness temperature before reionization

The slice-wise model has one clear weakness. Before reionization, when T_b is in absorption, it produces too much small-scale T_b power: $+24.8\sigma$ at $k=0.46\,{\rm Mpc}^{-1}$, against $+4.1\sigma$ for the 3-D rhizome and $+12.8\sigma$ for `fno` (Table 12). The excess is concentrated in quiet slices whose true small-scale power is tiny; the power summed over the cone is in fact slightly low (ratio 0.97). The working explanation is that the spin variable u is predicted slice by slice, and nothing in a per-pixel loss constrains its small-scale scatter within a slice, whereas the 3-D model predicts it jointly along the LOS.

Three fixes that address data, post-processing and context did not work; a fourth, in the loss, did (Table 13).

The power loss compares the transverse power spectrum of each predicted slice with the truth in eight logarithmic bins from $k=0.3\,{\rm Mpc}^{-1}$ to the grid Nyquist frequency,

$$\mathcal{L}_P=\lambda\left\langle\left(\log(P_{\rm pred}+\epsilon)-\log(P_{\rm true}+\epsilon)\right)^2\right\rangle_{\rm slices,\ bins},\qquad(21)$$

with $\lambda=0.03$, applied to the mean-subtracted normalized T_b. It brings the absorption-era excess from $+24.8\sigma$ to

Table 11. Slice-wise 2-D reconstruction against the 3-D models, $x_{\rm HI}$, 200 test cones, all trained on the same 1,600 cones.

Metric	2-D w128	**2-D w64**	3-D rhizome w48	`cnn_whno_es`
$\lvert\Delta\tau_e\rvert/\sigma_{\rm Planck}$	**0.011**	**0.011**	0.019	0.058
$\lvert\Delta z(\bar{x}=0.5)\rvert$	**0.011**	0.014	0.028	0.098
$x_{\rm HI}$ PDF W_1, active	**0.011**	0.013	0.021	0.030
b_1 pull	1.01	**0.85**	1.58	3.41
P_ε ratio (ideal 1)	**0.95**	0.94	0.89	0.77
bubble-size W_1 (Mpc)	0.59	**0.56**	0.81	1.39
power error, mid k	**0.062**	0.073	0.174	0.172
$1-r$, small k	**0.119**	0.120	0.199	0.181
edge H^1, ±2 Mpc	**0.109**	0.110	0.138	0.145
front width (Mpc; truth 2.65)	**4.03**	4.10	5.91	7.54

Table 12. Main results, 199 multi-field test cones (medians). "Slice-wise" is the multi-field slice-wise 2-D rhizome trained on all cones; "+ T_b power" adds the power term of Section 4.10. Best value per row in bold.

Metric	Slice-wise	+ T_b power	3-D rhizome C	`fno` `tbs+histtb`
$\lvert\Delta\tau_e\rvert/\sigma_{\rm Planck}$	0.010	**0.008**	0.012	0.076
$\lvert\Delta z(\bar{x}=0.5)\rvert$ / duration	**0.010 / 0.018**	**0.010 / 0.018**	0.011 / **0.018**	0.017 / 0.044
$x_{\rm HI}$ PDF W_1, active	**0.0095**	0.0098	0.0156	0.0271
b_1 pull	0.99	**0.93**	1.35	2.28
P_ε ratio (ideal 1)	**0.954**	–	0.945	0.793
bubble-size W_1, active (Mpc)	0.56	**0.48**	0.69	1.21
$x_{\rm HI}$ power error, small k	0.168	**0.163**	0.223	0.311
$1-r$, small k	**0.096**	0.102	0.151	0.174
edge H^1, ±2 Mpc	**0.106**	0.112	0.131	0.147
front width (Mpc; truth 2.77)	**3.89**	3.94	4.38	6.17
T_b PDF W_1, active (mK)	0.79	**0.77**	1.06	1.19
pre-reion. T_b power dev., $k=0.46$	$+24.8\sigma$	$\mathbf{+1.5\sigma}$	$+4.1\sigma$	$+12.8\sigma$

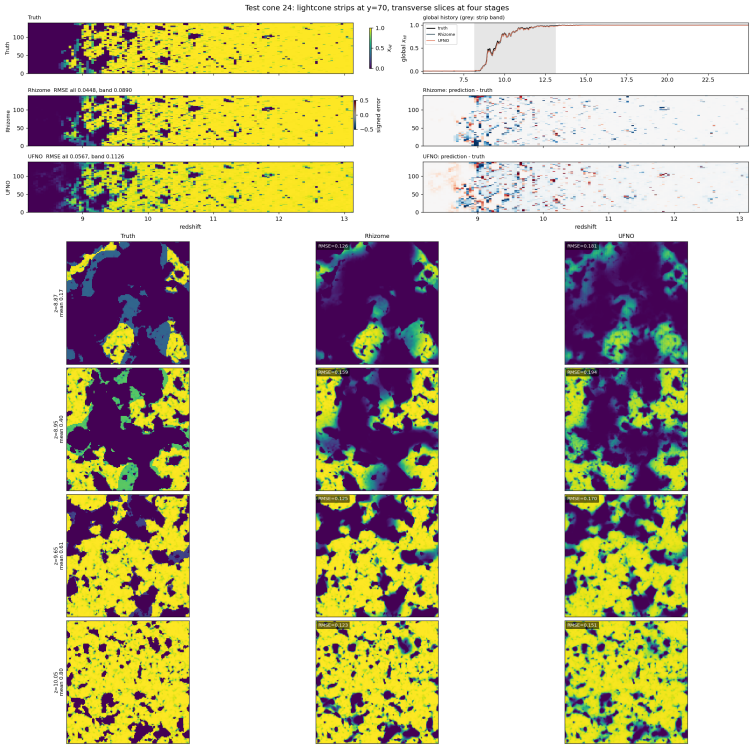


Figure 8. Test cone 24. *Top left:* $x_{\rm HI}$ lightcone strips for truth, rhizome (slice-wise multi-field) and the `fno` baseline (labelled UFNO). *Top right:* global history and signed error strips. *Bottom:* transverse slices at mean neutral fraction 0.17, 0.40, 0.61 and 0.80, with per-slice RMSE. Full-cone RMSE 0.045 (rhizome) against 0.057 (`fno`). The `fno` fronts are blurred, and it leaves a faint haze inside ionized regions.

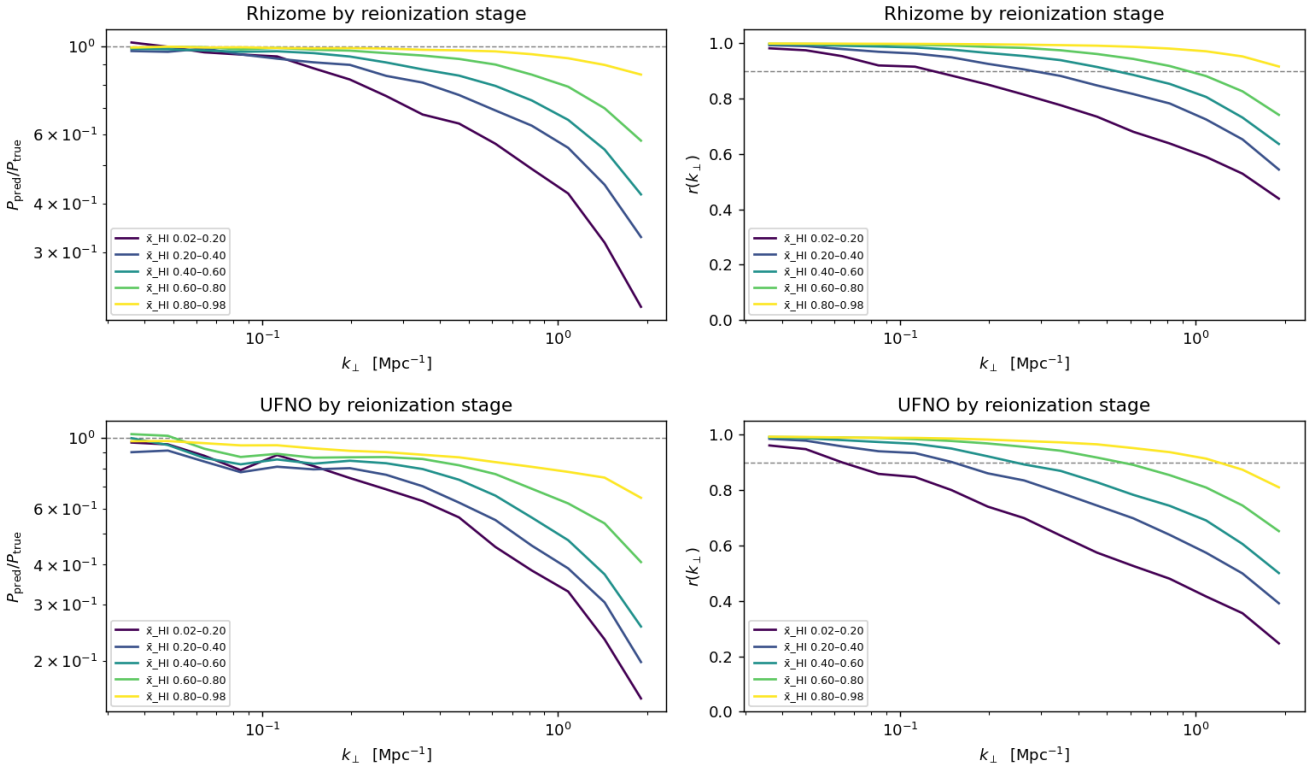


Figure 9. x_{HI} power ratio (left) and cross-correlation $r(k_\perp)$ (right) by reionization stage, multi-field slice-wise rhizome (top) and `fno tbs+histtb` (bottom), 199 test cones.

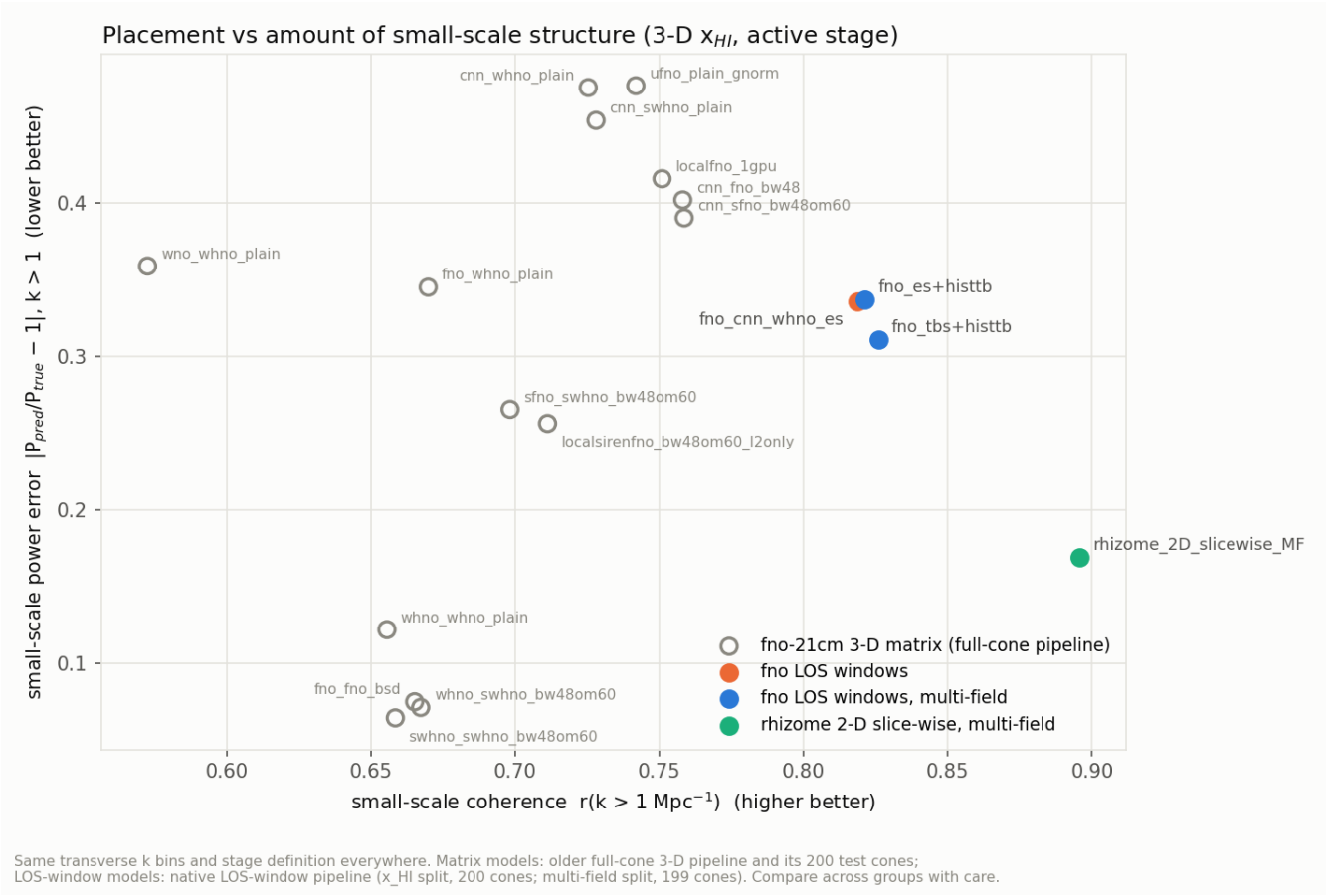


Figure 10. Small-scale coherence $r(k > 1\,\mathrm{Mpc}^{-1})$ against small-scale power error for the 3-D x_{HI} models of the earlier operator study (open circles), the `fno` LOS-window models (filled) and the multi-field slice-wise rhizome (green), active stage. Better models lie to the lower right. Models from different pipelines and splits are compared with care.

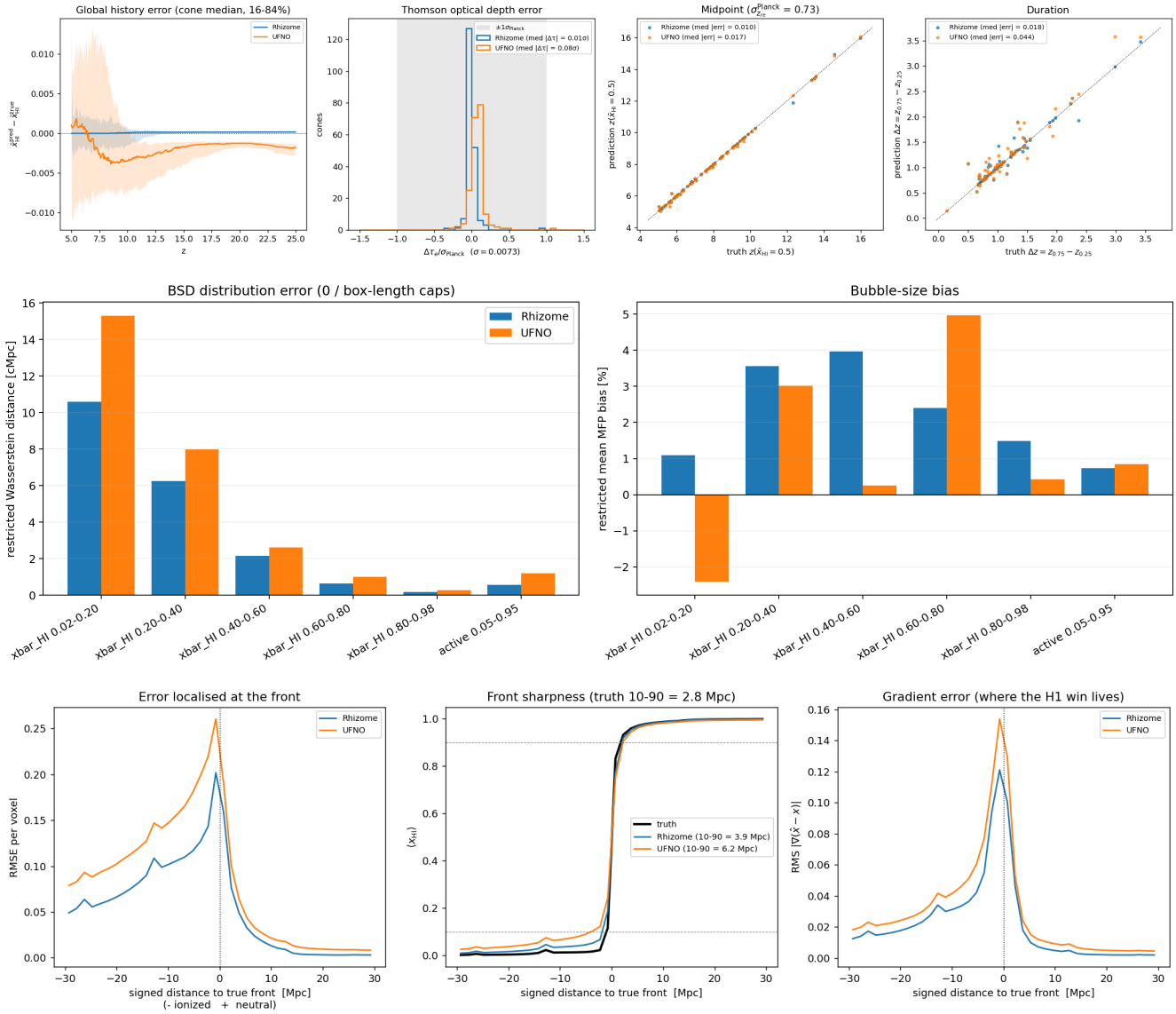


Figure 11. Physical metrics, rhizome (slice-wise multi-field) against `fno tbs+histtb`, 199 test cones. *Top:* global history error, τ_e error in Planck σ (median 0.01 against 0.08), and predicted against true midpoint and duration. *Middle:* bubble-size distributions and their W_1 distance to the truth by stage (active 0.56 against 1.21 Mpc). *Bottom:* error profiles across ionization fronts; front width 3.9 against 6.2 Mpc (truth 2.77).

Table 13. Attempts to remove the absorption-era T_b power excess of the slice-wise model (power deviation at $k = 0.46\,\mathrm{Mpc}^{-1}$, plain model: $+24.8\sigma$).

Attempt	Excess	Outcome
3.9× more training cones	$+24.8\sigma$	T_b values improve (W_1 −31%), excess unchanged: not a data limit
smooth u along the LOS ($\sigma = 4$ cells)	$+1.1\sigma$	removes the excess only by destroying real LOS structure: T_b RMSE +40–110%, summed power ×1.3; rejected
LOS context ±128 cells	$+36.0\sigma$	worse; per-voxel accuracy unchanged; not adopted
small-scale T_b power loss	$\mathbf{+1.5\sigma}$	fixes the excess at every scale; costs 4–8% per-voxel RMSE

$+1.5\sigma$, better than the 3-D rhizome, without overshooting the amplitude. It also improves the reionization-era statistics: T_b power within 1–2% at all scales, bubble-size W_1 −14% and τ_e −20% (Figure 13). The price is a 4–8% higher per-voxel RMSE, consistent over 80–90% of the cones, and slightly worse edges. This is the familiar trade-off between perceptual realism and pixel distortion: a per-pixel loss favours slightly smoothed fields, while the power term enforces realistic small-scale structure.

Applying the same term to $x_{\rm HI}$ as well roughly halves the high-k $x_{\rm HI}$ power deficit (from 0.163 to 0.089 over active slices; in the latest stage from 0.59 to 0.15), gives the closest value distribution of all models (W_1 0.0082), and keeps the absorption-era fix ($+1.3\sigma$). It costs another 2–4% in per-voxel RMSE, and the extra small-scale power is only partly in phase with the truth (small-scale incoherence +7%). The three variants are useful for different purposes: the plain model for per-voxel accuracy, the T_b power loss for 21 cm statistics, and both power terms for the most realistic small-scale structure in both fields.

A further third of the absorption-era T_b error is a constant offset per slice, of random sign from cone to cone. Its size (1.6–2.0 mK RMS) is nearly the same for all three model families and comparable to the error of the global-history emulator that provides the $\bar{T}_b(z)$ input (about 2.1 mK). It

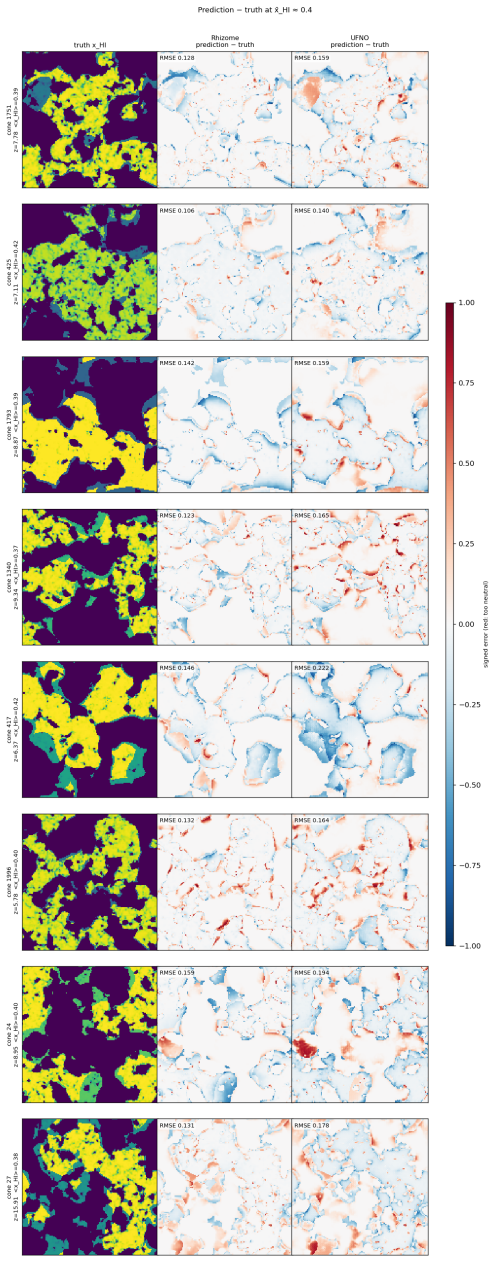


Figure 12. Signed x_{HI} error maps at $\bar{x}_{\mathrm{HI}} \approx 0.4$ on eight test cones (red: predicted too neutral), rhizome against `fno tbs+histtb`.

is therefore limited by the information in the input rather than by the operator, and the power loss, which acts only on fluctuations, leaves it unchanged.

4.11 Learned kernels: no spectral collapse

Neural networks tend to learn low frequencies first, and a Fourier layer can end up using only its lowest modes. Figure 14 tests this for the rhizome by comparing the RMS weight of each Fourier mode of the trained kernel with its value at initialization. The weights grew 13–45-fold almost evenly up to the band edge: the ratio of high-quarter to low-quarter weight is 0.78–0.94 for all rhizome models, against 0.10–0.45 for a U-FNO in an earlier study. Models with 36 or 48 modes also use their extra modes, so the weaker 48-mode result of the sweep reflects the short training budget, not unused capacity. A spectral-bias correction such as a SIREN-generated kernel would therefore not help here; full-budget runs with more modes are the cheaper next step.

4.12 Cost

Per training slice, the 2-D model is about four times slower than the 3-D models, since it runs full-resolution recurrent updates in float32. It needs 15–20 times fewer training slices, however (2.2 M against 33–49 M), so a complete training run is four to seven times cheaper: 3.5 h on one A100 for the best model, against about 18 h on an H200 for the 3-D rhizome and 44.7 h on an A100 for `cnn_whno_es`.

At inference, a full multi-field cone takes 5.5 s end to end on an A100 and 3.7 s on an H200 once the LOS bands are built on the GPU and the next data window is prefetched (Figure 15). This makes the slice-wise model the fastest model end to end, 1.8–2.3 times faster than both 3-D models; at that point inference is limited by reading the data. The GPU forward pass alone (3.8 s / 1.7 s) matches the 3-D rhizome and is about twice as slow as `fno`. Mixed precision is not worthwhile: bfloat16 gains only 5–14%, because the FFTs stay in float32, and it shifts x_{HI} by up to 0.08.

4.13 Ablations

Table 14 collects what each design and training choice was worth across the experiments above. Each line compares two otherwise identical single-seed runs.

One planned ablation has not yet been run: a controlled study of the gate mechanism itself. It compares the full rhizome with six otherwise identical arms (no gates, source-only, receiver-only, gates computed from the conditioning only, gates computed once from the initial state, and static per-channel gates), five seeds each, all retrained from scratch. Until then, the comparison with `fno` establishes that the rhizome *as a whole* works better on this task, but not how much of the gain is due to the state-dependent gating specifically, as opposed to the recurrence, the loss and the LOS input.

4.14 Discussion

What the rhizome does well. Across every configuration, the rhizome's advantage over the `fno` models is larger on physical observables than on per-voxel RMSE. It places ionization fronts more accurately (higher small-scale coherence), keeps them sharper, and reproduces bubble-size distributions, bias and stochasticity more faithfully. These are exactly the quantities that Section 1.5 identified as weak points of a purely spectral, stationary kernel. They are also the ones that matter for 21 cm inference, where bubble morphology and the field-level bias enter the observables.

What the line of sight needs. The most consequential finding is not about the operator itself. A 2-D model with ±32 cells of LOS context, applied slice by slice, outperforms native 3-D models trained on the same data, and does so at a fraction of the cost. Together with the periodic tiling of the lightcones, this suggests that the bottleneck of the 3-D models is how they handle the LOS, not their transverse expressivity.

Caveats. Several confounds remain. All results are single seeds, and differences of a few percent (for example A against A0) may be partly noise. Part of the advantage in τ_e and the global history comes from training with BCE rather

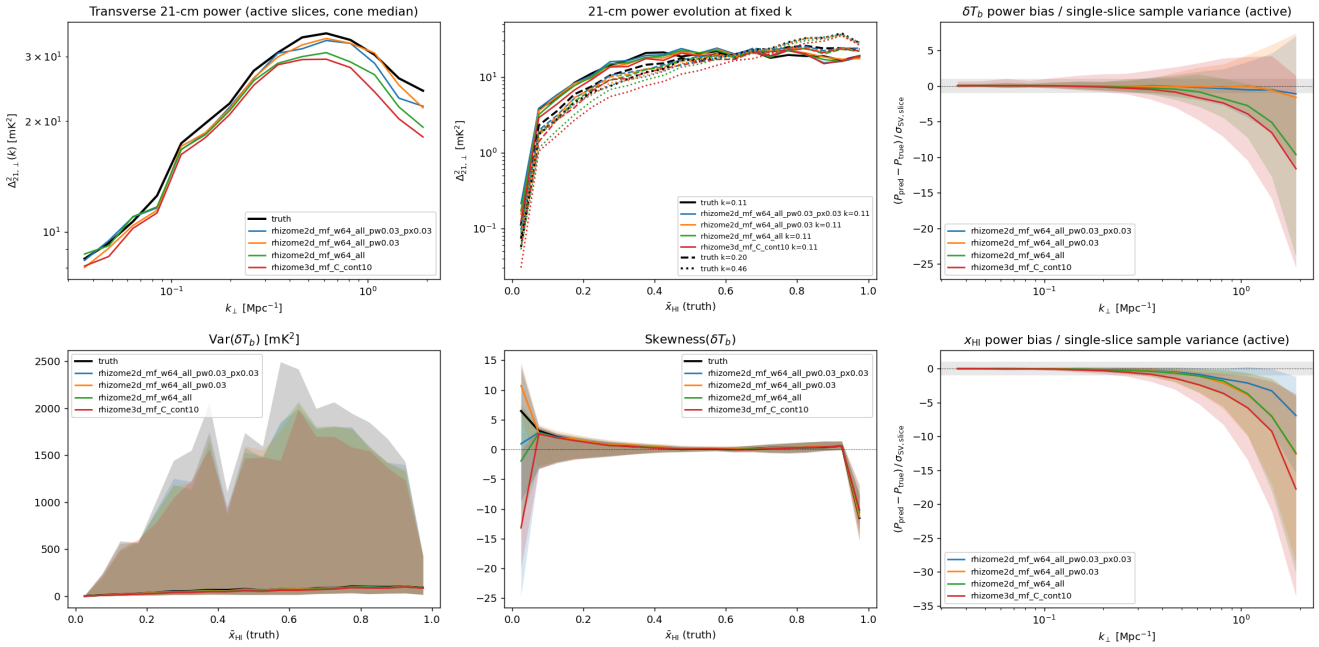


Figure 13. 21 cm statistics of the power-loss variants during reionization, 199 test cones: plain slice-wise model, + T_b power loss (`pw0.03`), + T_b and $x_{\rm HI}$ power loss (`pw0.03_px0.03`), and 3-D rhizome C. *Right column:* power bias in units of single-slice sample variance for T_b (top) and $x_{\rm HI}$ (bottom). Both power-loss models stay near zero to $k \approx 2\,\mathrm{Mpc}^{-1}$ in T_b; the plain model and 3-D C fall to -10 and -12σ.

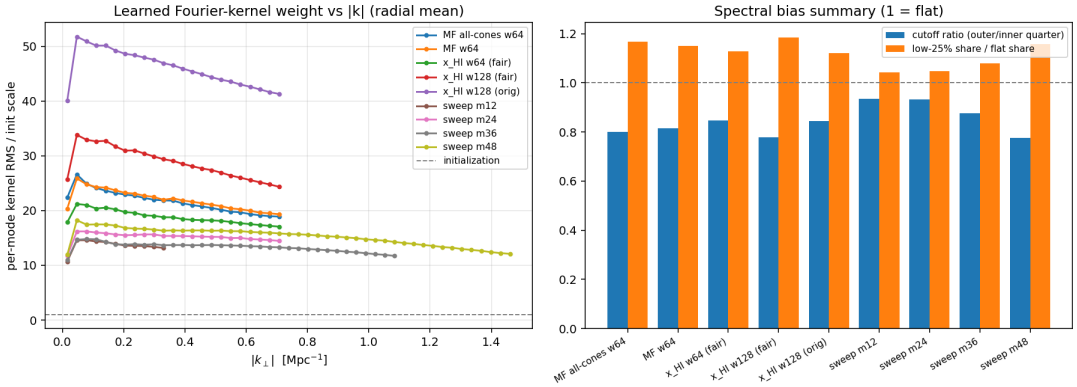


Figure 14. Per-mode RMS of the learned Fourier kernel relative to initialization. Radial profiles over wavenumber; they are flat to the band edge: there is no collapse onto low frequencies.

Table 14. Summary of ablations. Effects are relative changes in the stated metric; negative is better.

Change	Metric	Effect
LOS bands vs single slice (2-D)	mixed-slice RMSE	-36% (0.174 → 0.112)
width 64 → 128 (2-D)	mixed-slice RMSE	-3% at 4× parameters
width 48 → 64 (3-D)	test RMSE	$+6\%$ (worse)
lr $2\times10^{-3} \to 3\times10^{-4}$ (3-D)	test RMSE	$+13\%$ (worse)
warm restart (3-D)	test RMSE	-3 to -6%
history channels (3-D, A → B)	RMSE $x_{\rm HI}$ / T_b	-21% / -34%
structured T_b head (B → C)	T_b RMSE; timing	-5%; -15 to -30%
slice-wise 2-D vs 3-D (same data)	$1-r$; edge H^1	-40%; -21%
all 6,169 vs 1,594 training cones	median per-cone RMSE	-3.6% ($x_{\rm HI}$), -5.5% (T_b)
LOS context ±128 vs ±32	per-voxel RMSE	no significant change; duration -37%; absorption T_b worse
T_b power loss	pre-reion. T_b power dev.	$+24.8\sigma \to +1.5\sigma$; RMSE +4 to +8%
+ $x_{\rm HI}$ power loss	small-k $x_{\rm HI}$ power error	-45%; RMSE further +2 to +4%

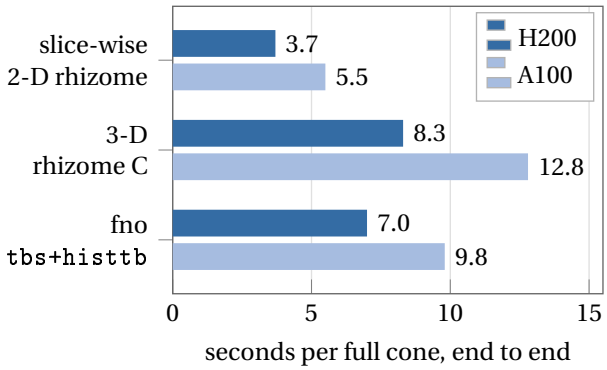


Figure 15. End-to-end inference time for one full multi-field test cone (x_{HI} and T_b), including data reading.

than from the operator, and the `fno` models should be retrained with BCE to separate the two. The contribution of the gating itself has not been isolated (Section 4.13). Finally, all results are at one resolution; the grid-independence of the parameters has not been put to the test.

A NOTATION

Symbol	Meaning
Operator learning	
f, g	input and output functions
$\mathcal{U}$, $\mathcal{V}$	input and output function spaces
D	spatial domain; N number of grid points
$\kappa(r, r', \theta)$	kernel of an integral operator
$\mathscr{F}$, $\mathscr{F}^{-1}$	Fourier transform and inverse
$R_\phi(k)$	learned Fourier coefficients of the kernel (FNO)
$k_{\max}$	number of retained Fourier modes
d_v	channel width
P, Q, W, σ	lifting, projection, pointwise linear path, nonlinearity
Rhizome operator	
$\boldsymbol{h}^t$, $\boldsymbol{c}$, $\boldsymbol{m}^t$	hidden state at update t, conditioning, message
A_θ, B_θ	receiver and source gates, values in $(0, 2)$
V_θ	value map
κ_θ, $R(k)$	base spatial kernel and its Fourier coefficients
κ_t^{eff}	effective, state-dependent kernel at update t
T, Δt	number of updates (6), step size (0.5)
21 cm physics and evaluation	
x_{HI}	neutral hydrogen fraction; $\bar{x}_{\mathrm{HI}}$ its global mean
T_b	21 cm brightness temperature (mK)
δ, z, ϑ	density contrast, redshift, simulation parameters
u	spin variable, $1 - e^u$ the spin-temperature factor
$T_0(z)$, F_v	T_b amplitude and velocity-gradient factor
τ_e	CMB optical depth to reionization
$r(k)$	cross-correlation coefficient of prediction and truth
b_1, P_ε	linear bias and stochastic power of the EFT expansion
W_1	Wasserstein-1 distance
LOS	line of sight

REFERENCES

[1] Shun-ichi Amari. Dynamics of pattern formation in lateral-inhibition type neural fields. *Biological Cybernetics*, 27(2): 77–87, 1977.

[2] Francesca Bartolucci, Emmanuel de Bézenac, Bogdan Raonić, Roberto Molinaro, Siddhartha Mishra, and Rima Alaifari. Representation equivalent neural operators: a framework for alias-free operator learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[3] Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

[4] Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are RNNs: Fast autoregressive transformers with linear attention. In *International Conference on Machine Learning (ICML)*, pages 5156–5165, 2020.

[5] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Neural operator: Graph kernel network for partial differential equations. *arXiv preprint arXiv:2003.03485*, 2020.

[6] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations (ICLR)*, 2021.

[7] Andrei Mesinger, Steven Furlanetto, and Renyue Cen. 21cmfast: a fast, seminumerical simulation of the high-redshift 21-cm signal. *Monthly Notices of the Royal Astronomical Society*, 411(2):955–972, 2011.

[8] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 234–241, 2015.

[9] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

[10] Gege Wen, Zongyi Li, Kamyar Azizzadenesheli, Anima Anandkumar, and Sally M. Benson. U-fno—an enhanced fourier neural operator-based deep-learning model for multiphase flow. *Advances in Water Resources*, 163:104180, 2022.